

Over The Air Computing

Master Thesis

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Abstract – Over-the-Air (OTA) is an emerging technique that leverages the superposition property of wireless channels to perform computations directly during the transmission process. This approach has the potential to significantly reduce latency and energy consumption in distributed systems, making it particularly suitable for applications in Internet of Things (Internet of Things (IoT)) networks, edge computing, and real-time data processing. This thesis explores the theoretical foundations of OTA, including channel modeling, signal processing techniques, and error analysis. Practical implementations and case studies demonstrating the effectiveness of OTA in various scenarios are also presented.

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List of Acronyms and Abbreviations

AGC	Automatic Gain Control 18
AM	Amplitude Modulation 5, 7, 9, 14
AWGN	Additive White Gaussian Noise 5, 7, 9, 15, 18, 26, 39
BPSK	Binary Phase-Shift Keying 14
CFO	Carrier Frequency Offset 31, 32, 34–36
CSI	Channel State Information 30
DSP	Digital Signal Processing 28
ED	End Device 5, 18, 22
FC	Fusion Center 6–8, 13, 17, 19, 20, 22, 30, 31, 34
FIR	Finite Impulse Response 17, 18
IoT	Internet of Things 1
ISI	Inter-Symbol Interference 8, 12, 18, 20, 22, 35, 37, 38
LO	Local Oscillator 32
MAC	Multiple Access Channel 5, 6, 15
MSE	Mean Squared Error 8, 10, 12, 15, 18, 20, 21, 25, 28, 32, 36
OFDM	Orthogonal Frequency-Division Multiplexing 5
OTA	Over-the-Air 1, 4–6, 8, 18, 20, 30, 34, 37, 39, 40
PAM	Pulse Amplitude Modulation 6, 19, 34
PLL	Phase-Locked Loop 32
QAM	Quadrature Amplitude Modulation 14
RC	Raised Cosine 7, 19–21
RRC	Root Raised Cosine 7, 9, 10, 12–15, 17–19, 22, 25, 26, 31
SDR	Software Defined Radio 4, 5, 17, 39
SNR	Signal-to-Noise Ratio 10, 12, 21, 25, 28
sps	Symbols per Sample 10
TDMA	Time-Division Multiple Access 4, 5, 40

Introduction

Over-the-Air (OTA) Computing represents a paradigm shift in how distributed computations are executed across wireless networks. By exploiting the superposition property of the wireless Multiple-Access Channel (MAC), OTA allows simultaneous transmissions from multiple sensing nodes to be aggregated in the air, directly yielding a desired nomographic function of the distributed data. This approach inherently couples communication and computation, breaking the separation principle to achieve unprecedented spectral efficiency and reduced latency. These properties are increasingly critical for rapidly scaling applications such as Federated Learning (FL), distributed consensus, and real-time sensor analytics.

0.a Background and Motivation

Traditional orthogonal access schemes, such as Time-Division Multiple Access (Time-Division Multiple Access (TDMA)), suffer from severe scalability bottlenecks because their resource requirements scale linearly with the number of transmitting nodes [9]. In computation-oriented networks where the receiver is interested only in a function of the data (e.g., the arithmetic mean) rather than the individual messages themselves, orthogonal separation proves highly suboptimal [6], [7]. OTA computation leverages analog signal superposition to compute these functions directly at the physical layer, achieving a computation rate that is theoretically independent of the network size [5]. However, achieving this theoretical efficiency in practice requires precise control over the transmitted waveforms and rigorous synchronization across spatially distributed transmitters.

0.b Problem Statement and Research Objectives

While the theoretical foundations of computation over MAC are well-established [4], [7], translating these scaling laws into practical realizations remains a formidable challenge. The primary focus of this thesis is to bridge the gap between idealized OTA theory and physical deployment. By systematically implementing OTA architectures across multiple abstraction layers—ranging from mathematical formulations and discrete-time MATLAB simulations to dynamical Simulink models and ultimately Software-Defined Radio (SDR) platforms—this work aims to quantify how different modeling fidelities capture the impact of physical channel impairments. Specifically, the objectives are to:

- Establish a rigorous ideal OTA baseline and validate its underlying physics.
- Systematically model and quantify the degradation caused by severe channel impairments, particularly multipath fractional delay and asynchronous Carrier Frequency Offsets (CFO).
- Evaluate established filtering and equalization techniques (such as squaring loops, Costas loops, and non-coherent envelope detection) to combat these uncoordinated impairments.
- Lay the groundwork for a robust Software Defined Radio (SDR) implementation using GNU Radio that addresses hardware-level synchronization limits.

0.c Thesis Contributions and Organization

This thesis contributes a comprehensive, cross-platform methodology for evaluating OTA computation under non-ideal physical conditions. The structure of the thesis is organized as follows:

- **Ideal Over-the-Air Computation Baseline:** Establishes the fundamental physical and mathematical models, exploring Double-Sideband Amplitude Modulation, matched filtering, and the theoretical Mean Squared Error (MSE) scaling under pure Additive White Gaussian Noise (AWGN).

- **Multipath Propagation, Fractional Delay, and Intersymbol Interference:** Dissects the vulnerability of OTA computation to multipath propagation and asynchronous CFO, providing detailed mathematical derivations, discrete-time simulations, and continuous-time Simulink block-level architectures that expose the mechanics of intersymbol interference and phase misalignment.
- **Experimental Implementation** (forthcoming): Transitions the modeled architectures into hardware, detailing the GNU Radio flowgraphs and the practical hurdles of SDR deployment.
- **Experimental Results and Discussion:** Provides a comparative analysis of the simulation models and equalization strategies, alongside an evaluation framework for the experimental SDR findings.
- **Conclusion and Future Work:** Synthesizes the findings and proposes advanced adaptive equalization schemes, such as Unscented Kalman Filtering, for future exploration.

1 Ideal Over-the-Air Computation Baseline

Before introducing the practical impairments inherent to physical wireless channels—such as multipath delay spread, intersymbol interference, carrier frequency offsets, and time synchronization errors—it is essential to establish an analytical and simulation baseline of OTA computation under ideal conditions. The foundational premise of OTA computing is its ability to overcome the communication bottleneck of distributed wireless sensor aggregation. In conventional orthogonal multi-access protocols (e.g., TDMA or Orthogonal Frequency-Division Multiplexing (OFDM)), the time required to collect readings from K distributed End Device (ED)s scales linearly with the network size, incurring a latency complexity of $\mathcal{O}(K)$. By contrast, OTA computing exploits the superposition property of the wireless Multiple Access Channel (MAC) to execute nomographic function aggregation directly across the physical propagation medium in a single time slot:

$$\mathcal{O}(K) \rightarrow \mathcal{O}(1) \quad (1)$$

In this idealized baseline scenario, all transmitting nodes are assumed to be perfectly synchronized in carrier frequency, phase, and symbol timing. The propagation channel is modeled as a non-dispersive, flat-fading medium corrupted solely by Additive White Gaussian Noise (AWGN). This chapter presents an end-to-end multi-tiered experimental workflow that demonstrates this unimpaired baseline:

- (i) The physical principles and electrical circuit equivalent of electromagnetic superposition;
- (ii) The continuous-time and passband Amplitude Modulation (AM) mathematical formulation;
- (iii) Algorithmic discrete-time MATLAB modeling and multi-rate pulse-shaping code;
- (iv) Mean squared error and constellation diagnostic evaluations;
- (v) Analog eye diagram analysis comparing MATLAB and Simulink dynamics;
- (vi) Multi-rate block-level Simulink dynamical modeling; and
- (vii) Software-Defined Radio (SDR) realization and polyphase filter calibration in GNU Radio.

Establishing this rigorous benchmark provides the mathematical reference and experimental framework repeated throughout subsequent chapters to evaluate model impurities and channel imperfections.

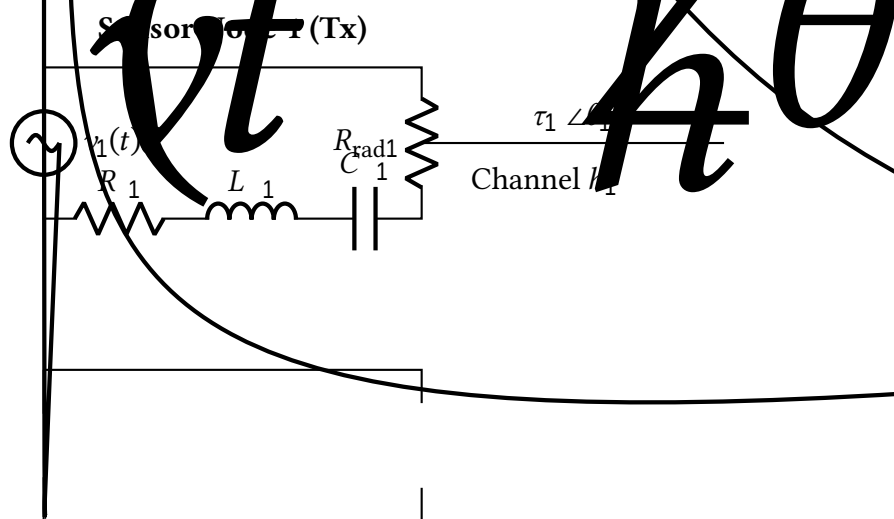
1.a Physical Principles and Electrical Circuit Equivalent of Electromagnetic Superposition

In conventional digital signal processors, arithmetic computation is executed through Boolean logic gates constructed from semiconductor transistors. In OTA computation, however, arithmetic addition is performed natively by free space through Maxwell's electrodynamic equations.

When K distributed sensor nodes concurrently radiate radio-frequency currents through their respective antennas, the emitted electric field vectors $\mathbf{E}(\mathbf{r}, t)$ superimpose linearly in the spatial continuum according to the principle of electromagnetic superposition:

$$\mathbf{E}_{\text{total}}(\mathbf{r}, t) = \sum_{i=1}^K \mathbf{E}_i(\mathbf{r}, t) \quad (2)$$

At the receive antenna aperture of the central Fusion Center (FC), the incident electric field induces an open-circuit terminal voltage proportional to the spatial integral of the field weighted by the antenna's effective vector length.



To ensure finite spectral occupancy and satisfy regulatory bandwidth constraints, each transmitting device shapes its symbol sequence with a Root Raised Cosine (RRC) pulse $p(t)$ having roll-off factor $\alpha \in [0, 1]$ and symbol period T . The pulse satisfies unit-energy normalization:

$$\int_{-\infty}^{\infty} p^2(t) dt = 1 \quad (3)$$

The continuous-time baseband waveform radiated by sensor k is given by:

$$s_k(t) = \sum_{m=-\infty}^{\infty} x_k[m] p(t - mT) \quad (4)$$

1.b.i Passband Double-Sideband Amplitude Modulation

In practical radio frequency systems, baseband waveforms are upconverted to a carrier frequency

Because the Nyquist pulse satisfies the zero-Inter-Symbol Interference (ISI) property ($r(0) = 1$ and $r(lT) = 0$ for all non-zero integers $l \neq 0$), the intersymbol cross-terms vanish identically:

$$y[n] = \sum_{=1} x[n] + \tilde{w}[n] \quad (11)$$

where $\tilde{w}[n] \sim \mathcal{N}(0, \sigma^2)$. To compute the desired arithmetic average of the K sensor readings, the FC applies an unweighted post-processing scaling factor of $1/K$:

$$\hat{\mu}[n] = \frac{1}{K} y[n] = \frac{1}{K} \sum_{=1} x[n] + \frac{\tilde{w}[n]}{K} \quad (12)$$

1.b.iii Theoretical Mean Squared Error Scaling

The instantaneous estimation error at symbol index n is defined as the deviation between the reconstructed sample mean and the true arithmetic average:

$$e[n] = \hat{\mu}[n] - \frac{1}{K} \sum_{=1} x[n] = \frac{\tilde{w}[n]}{K} \quad (13)$$

The statistical Mean Squared Error (MSE) of the ideal Over-the-Air computation system is obtained by taking the mathematical expectation:

$$\text{MSE} = \mathbb{E}[|e[n]|^2] = \mathbb{E}\left[\left|\frac{\tilde{w}[n]}{K}\right|^2\right] = \frac{\sigma^2}{K^2} \quad (14)$$

Eq. (14) reveals the fundamental cooperative advantage of OTA aggregation: while individual sensor signals add coherently, thermal noise adds non-coherently, resulting in an inverse-quadratic noise suppression factor of $1/K^2$ relative to the unweighted channel noise variance [1], [9].

1.b.iv Distributive and Linear Time-Invariant Equivalence

A vital property of the ideal baseline concerns the mathematical equivalence between synthesizing each sensor's modulated waveform individually versus modulating the aggregated sequence directly. Let $u(t) = \sum x[m]\delta(t - mT)$ represent the ideal impulse train of sensor k . In the physical network, each sensor performs pulse shaping and passband mixing locally prior to transmission:

$$r_{\text{agg}}(t) = \sum_{=1} \left((u(t) * p(t)) \cos(2\pi f t) \right) \quad (15)$$

Because all sensors share an identical carrier frequency and phase, the carrier waveform factors out of the sum:

$$r_{\text{agg}}(t) = \left(\sum_{=1} (u(t) * p(t)) \right) \cos(2\pi f t) \quad (16)$$

Furthermore, convolution is a linear operator that distributes over addition:

$$\sum_{=1} (u(t) * p(t)) = \left(\sum_{=1} u(t) \right) * p(t) \quad (17)$$

Substituting Eq. (17) into Eq. (16) yields the optimized equivalent representation:

$$r_{\text{agg}}(t) = \left(\left[\sum_{=1} u(t) \right] * p(t) \right) \cos(2\pi f t) \quad (18)$$

This proof demonstrates that under ideal synchronization and identical channel gains, the order of operations commutes completely. This property enables substantial computational acceleration in discrete-time simulations: the algorithmic complexity scales from $\mathcal{O}(K \cdot N \cdot \text{sps} \cdot L_{\text{filt}})$ down to $\mathcal{O}(N \cdot \text{sps} \cdot L_{\text{filt}})$, achieving a K -fold reduction in runtime and memory allocation.

1.c Algorithmic MATLAB Implementation and Code Listings

To evaluate these theoretical principles numerically, two distinct discrete-time simulation architectures are modeled in MATLAB:

1. An unmodulated, single-sample-per-symbol baseline that isolates the pure mathematical summation under AWGN (Listing 1); and
2. A physically complete multi-rate pipeline incorporating upsampling, matched RRC pulse shaping, Double-Sideband AM, and coherent demodulation (Listing 2).

Listing 1: MATLAB script for ideal unmodulated OTA computation baseline.

```

1      % Ideal OTA Computation Simulation - Unmodulated Baseline
2      K = 20;                % Number of sensor nodes
3      N = 4000;             % Number of symbols
4      SNR_dB = 20;          % Signal-to-Noise Ratio (dB)
5
6      % Generate agricultural temperature readings around 27 deg C
7      X = 27 + randn(K, N);
8
9      % Over-The-Air superposition (Physical MAC Adder)
10     Ideal_Sum = sum(X, 1);
11
12     % Add AWGN according to measured average signal power
13     Received_Signal = awgn(Ideal_Sum, SNR_dB, 'measured');
14
15     % Fusion Center arithmetic average calculation
16     Estimated_Average = Received_Signal / K;
17
18     % Empirical Mean Squared Error computation
19     MSE = immse(Estimated_Average, mean(X, 1));
20     fprintf('Ideal Baseline MSE at %d dB SNR: %e\n', SNR_dB, MSE);

```

While Listing 1 demonstrates the fundamental principle in its abstract discrete-time form, practical wireless transceivers mandate bandlimiting and RF upconversion. Listing 2 presents the complete multi-rate architecture. The discrete symbol sequence is upsampled by an integer factor `sps_val`, convolved with a unit-energy normalized RRC filter, modulated onto carrier frequency $F = F_c/4$, corrupted by channel noise, coherently demodulated with a matched RRC filter, and downsampled at the calibrated strobe instant.

Listing 2: MATLAB multi-rate script for OTA computation with matched RRC filtering and coherent AM.

```

1      % Multi-Rate OTA Computation with Matched RRC and AM
2      txUpSampled = upsample(X', sps_val)'; % Dimension: K x (N * sps)
3

```

```

4      % Design Root Raised Cosine filter with unit energy normalization
5
6
7      % Commutative superposition: pulse-shape aggregate impulse sequence
8      agg_impulses = sum(txUpSampled, 1);
9      tx(basebandbandwidth, agg_impulses, rcFilter, 'same');
10
11     % Double-Sideband Amplitude Modulation (Fc = Modulation
12     t = 0:length(ex_baseband)-1) / Fs;
13     aseband    conv(agg_il  0

```

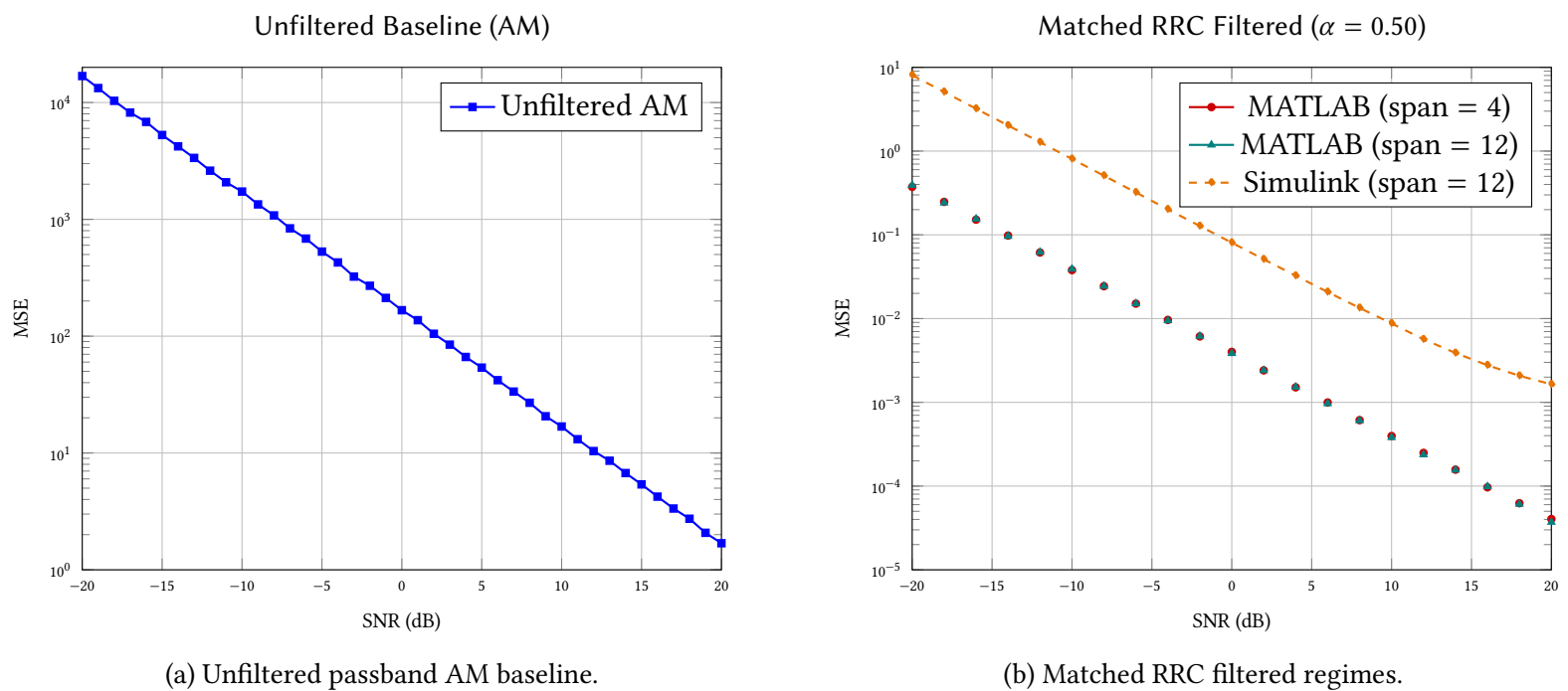


Figure 2: Two-dimensional MSE comparison between the unfiltered baseline and the matched RRC filtered systems across SNR (-20 dB to $+20$ dB). Both configurations exhibit log-linear error decay without an error floor, confirming pure AWGN limitation.

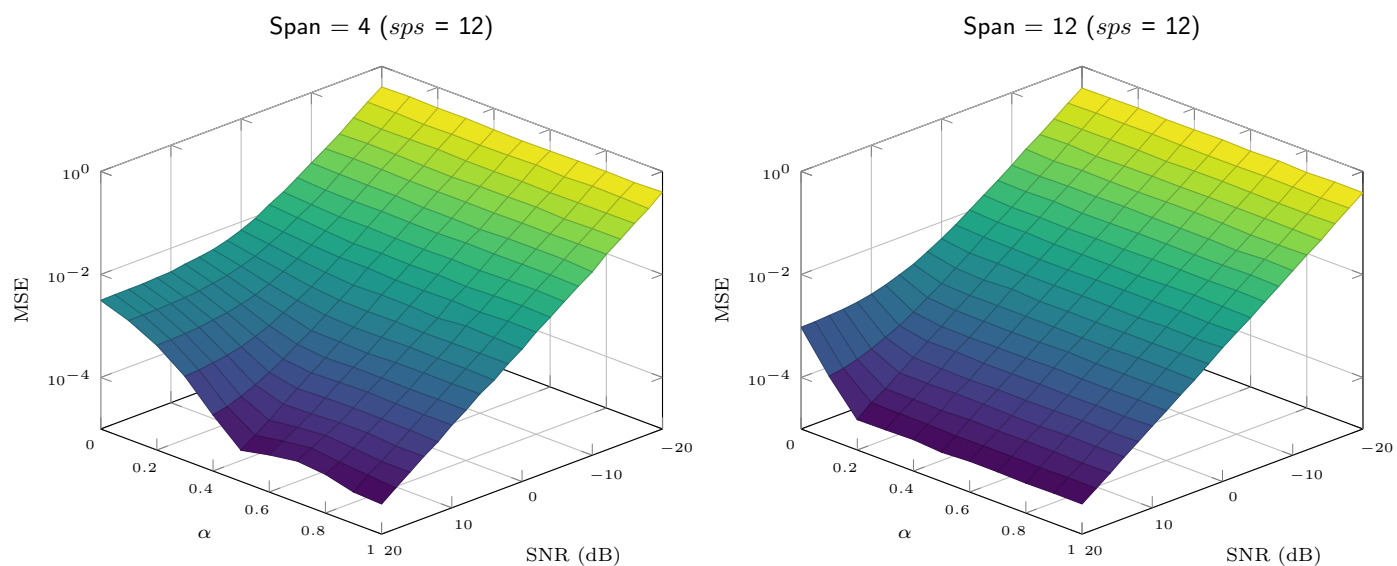


Figure 3: Synchronous 3D surface animation of the Over-The-Air MSE across roll-off factor $\alpha \in [0.0, 1.0]$ and SNR (-20 dB to $+20$ dB) under oversampling scaling ($sps \in \{12, 24, 48, 96\}$) for filter span 4 (left) versus span 12 (right).

low roll-off factors ($\alpha \rightarrow 0$).

- **Span 12 (Right):** Extending the filter memory to 12 symbol durations allows the RRC tails to decay naturally below -40 dB. Consequently, the resulting MSE surface is exceptionally flat across all values of α , demonstrating that truncation-induced ISI is completely eliminated.
- **Oversampling Convergence:** Scaling the oversampling ratio from $\text{sps} = 12$ to $\text{sps} = 96$ yields negligible change in the MSE surface. This confirms that once the Nyquist sampling criterion is satisfied ($F > 2B$), additional oversampling provides diminishing returns under ideal synchronization.

1.e Constellation Diagnostics and Variance Discrepancy Mechanics

To examine the structural integrity of the recovered signal at the symbol decision instant, constellation scatter plots are evaluated at a reference SNR of 20 dB.

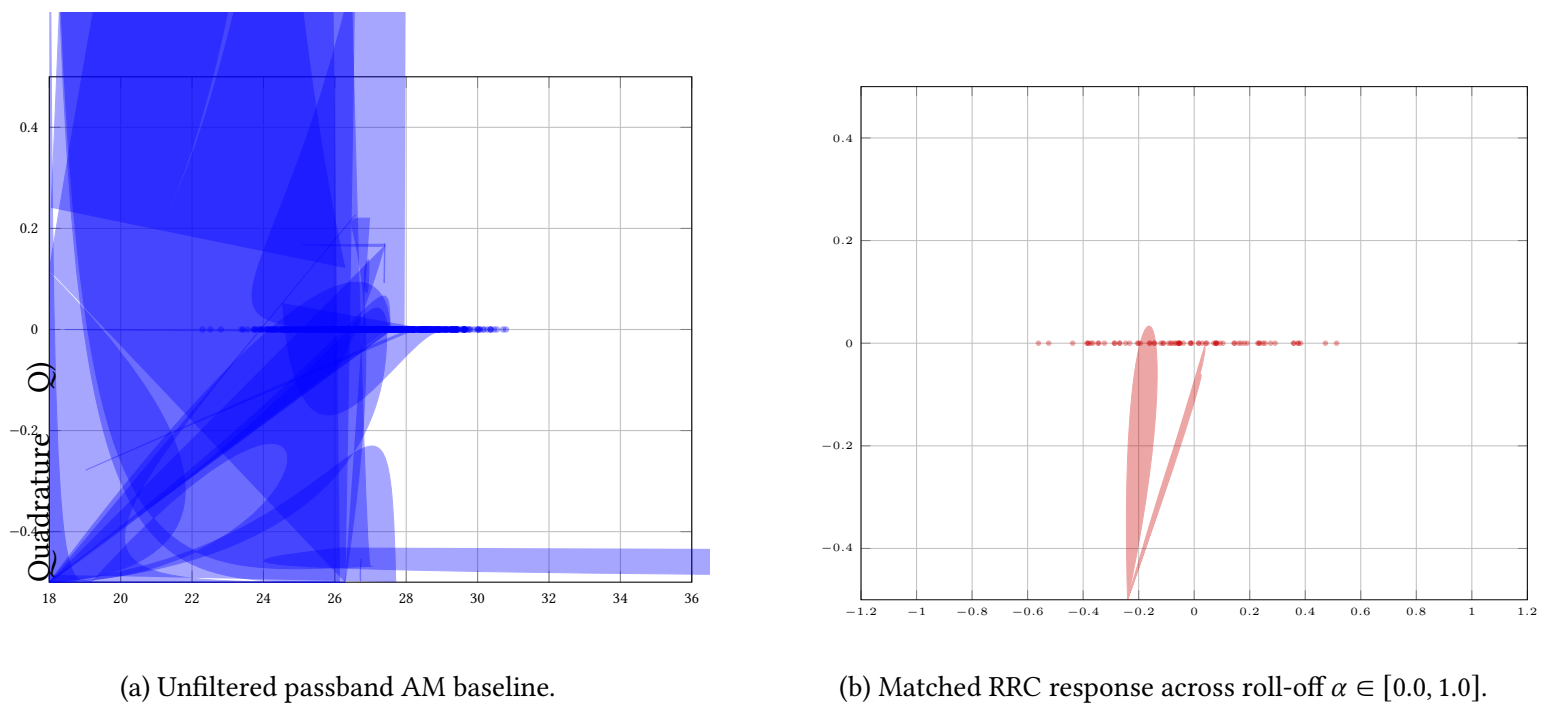


Figure 4: Constellation diagram comparison at SNR = 20 dB. Left: Unfiltered AWGN baseline exhibiting wider dispersion ($I \in [22, 32]$). Right: Animated matched RRC filter response exhibiting tight residual clustering ($I \in [-0.8, 0.8]$) across roll-off α (span = 12, $\text{sps} = 48$).

A prominent empirical observation in Figure 4 concerns the marked difference in constellation dispersion: in the unfiltered scenario (Figure 4a), the received values scatter across a broad interval spanning $I \in [22.3, 31.9]$ (with extreme samples reaching 17 to 38 in unmodulated tests). In contrast, under matched RRC filtering (Figure 4b), the points are tightly clustered within a narrow band of $I \in [-0.73, 0.73]$.

This apparent paradox—wherein an unfiltered channel theoretically free of pulse-shaping distortion exhibits *greater* variance than a filtered channel—is governed by two distinct physical and simulation mechanisms:

1.e.i Physical Mechanism: Matched Filtering and Noise Bandwidth Rejection

In communication theory, an unfiltered receiver possesses an unconstrained noise equivalent bandwidth equal to the entire sampling rate ($B = F/2$). Consequently, it admits thermal noise energy across the entire frequency spectrum. By contrast, the Root Raised Cosine matched filter at the receiver acts as a strictly bandlimited filter that passes only the signal bandwidth $B = \frac{1+}{2s}$, completely rejecting out-of-band noise. By the North-Turin matched filter theorem, convolving the received noisy waveform with $p(-t)$ executes coherent integration: signal components add constructively, while zero-mean noise samples partially cancel, providing an effective processing gain proportional to the oversampling factor.

1.e.ii Simulation Mechanism: Interaction of AWGN with Sensor DC Pedestal

The numerical discrepancy in MATLAB's `awgn( . . . , 'measured' )` function originates from the non-zero agricultural temperature mean (27 °C). With $K = 20$ sensors, the instantaneous aggregate baseline is:

$$\text{Ideal_Sum}[n] \approx K \times 27 = 540 \quad (19)$$

In the unmodulated script (Listing 1), MATLAB measures the total average signal power:

$$P_{\text{signal}} = \frac{1}{N} \sum_{n=1}^N (\text{Ideal_Sum}[n])^2 \approx 540^2 = 291,600 \quad (20)$$

At SNR = 20 dB (a power ratio of 100 : 1), MATLAB injects noise with variance:

$$\sigma_{\text{noise}}^2 = \frac{P_{\text{signal}}}{100} = \frac{291,600}{100} = 2,916 \implies \sigma_{\text{noise}} = \sqrt{2,916} \approx 54 \quad (21)$$

The FC divides the noisy sum by $K = 20$:

$$\hat{\mu} = \frac{540 + w}{20} = 27 + \frac{w}{20} \quad (22)$$

The standard deviation of this estimate is:

$$\sigma_{\text{est}} = \frac{\sigma_{\text{noise}}}{K} = \frac{54}{20} = 2.7 \quad (23)$$

For a Gaussian distribution with $\sigma_{\text{est}} = 2.7$, the 3σ (99.7%) confidence boundary is:

$$27 \pm 3(2.7) = 27 \pm 8.1 = [18.9, 35.1] \quad (24)$$

which precisely accounts for the wide scatter observed in Figure 4a.

Conversely, in the multi-rate RRC pipeline, the sequence is upsampled by `sps_val = 10`. The impulse train contains non-zero values in only 1 out of every 10 samples. Because the RRC filter is normalized to unit energy ($\sum g[n]^2 = 1$), the pulse energy is distributed over 10 samples, reducing the measured average continuous power by a factor of `sps`:

$$P_{\text{measured}} \approx \frac{(K \cdot 27)^2}{\text{sps}} = \frac{291,600}{10} = 29,160 \quad (25)$$

Consequently, MATLAB injects a tenfold smaller noise variance:

$$\sigma^2 = \frac{29,160}{100} = 291.6 \quad (26)$$

Passing through the receiver matched filter preserves this unit-energy variance:

$$\sigma_{\text{filtered}}^2 = \sigma^2 \sum g^2[n] = 291.6 \times 1 = 291.6 \quad (27)$$

Dividing by $K = 20$ yields an estimate standard deviation of:

$$\sigma_{\text{est, RRC}} = \frac{\sqrt{291.6}}{20} \approx 0.85 \quad (28)$$

The resulting 3σ interval is $27 \pm 3(0.85) = [24.45, 29.55]$, yielding an exceptionally tight distribution. Furthermore, when zero-mean residual modulation is employed ($\Delta x = x - 27$), the DC pedestal is removed entirely, centering the constellation at $I = 0$ with $\sigma_{\text{est}} < 0.25$, as shown in Figure 4b.

1.f Eye Diagram Diagnostics: MATLAB and Simulink Comparative Analysis

In conventional digital communications (e.g., Binary Phase-Shift Keying (BPSK) or Quadrature Amplitude Modulation (QAM)), eye diagrams exhibit an open, diamond-shaped central aperture delimiting discrete decision thresholds (± 1). In uncoded analog Over-the-Air computation, however, the eye diagram exhibits a fundamentally different topology that must be interpreted as an analog signal integrity and timing sensitivity diagnostic:

1. **Continuous Amplitude Band:** Because sensor readings are continuous random variables ($X \sim \mathcal{N}(27, 1)$), the aggregated amplitude occupies a continuum of real values without forbidden intermediate regions, forming a smooth flowing ribbon.
2. **Nyquist Strobe Verification:** In an ideal Nyquist system, the first time derivative of every pulse passes through zero at the nominal sampling instants:

$$\left. \frac{dp(t)}{dt} \right|_{=0} = 0 \quad (29)$$

At integer symbol strobe instants ($t/T \in \{1, 2, \dots\}$), the trajectory curves reach local extrema or inflection points, yielding perfectly horizontal slopes that confirm zero intersymbol leakage.

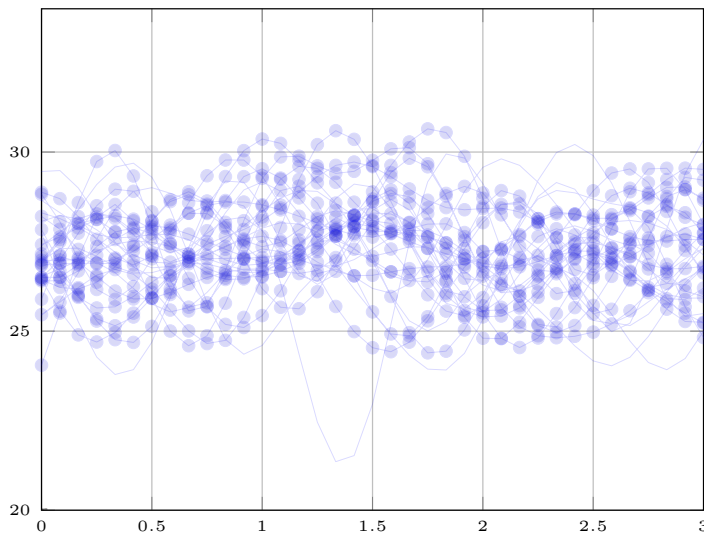
3. **Jitter and Timing Sensitivity:** Timing margin is quantified by the second derivative (curvature) around the strobe point:

$$y(T + \Delta t) \approx y(T) + \frac{1}{2} y''(T) \Delta t^2 \quad (30)$$

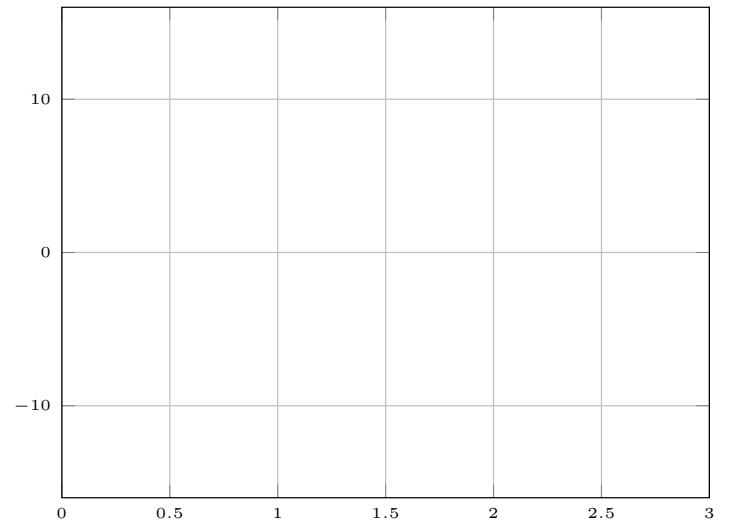
A flatter trajectory bundle around the strobe instant guarantees high robustness against transceiver clock jitter.

1.f.i MATLAB Eye Diagram Evaluation

Figure 5 compares the continuous-time passband AM eye diagram against the multi-rate matched RRC filter response across oversampling factors $sps \in \{12, 24, 48, 96\}$.



(a) Passband AM eye across $sps \in \{12, 24, 48, 96\}$.



(b) Matched RRC eye across $sps \in \{12, 24, 48, 96\}$ (span = 12, $\alpha = 0.70$).

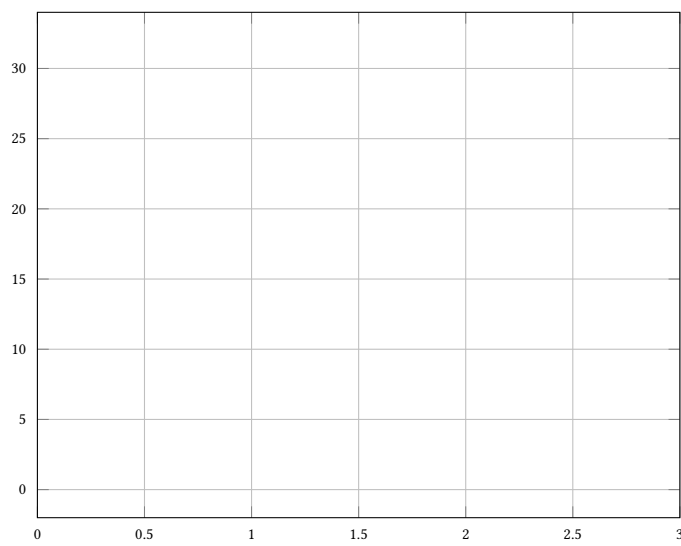
Figure 5: MATLAB eye diagram diagnostics. Left: Demodulated passband AM eye animation across oversampling factor $sps \in \{12, 24, 48, 96\}$, illustrating continuous-time trajectories oscillating about the 27 C mean without intersymbol zero-crossings. Right: Matched RRC filter eye animation (span = 12, roll-off $\alpha = 0.70$) across identical oversampling ratios, exhibiting zero-mean clustering and sharp nodal convergence at integer strobe instants.

In Figure 5a, the demodulated passband AM eye animation demonstrates how increasing the oversampling ratio resolves

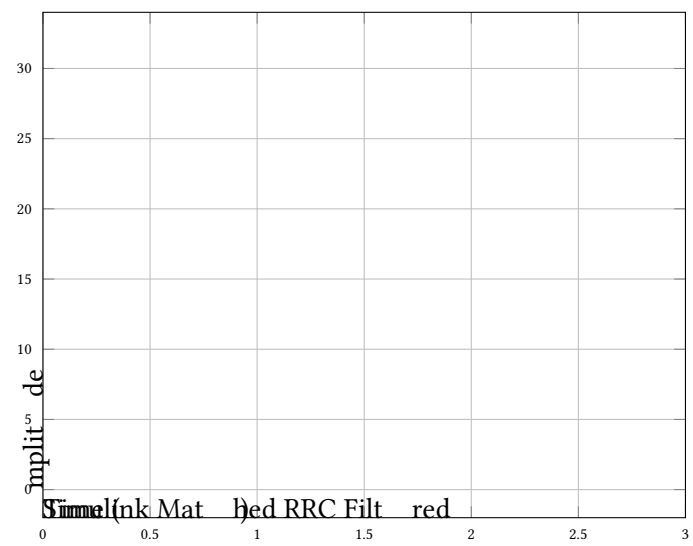
the continuous-time carrier and pulse transitions with progressive fidelity around the nominal mean of 27 C. Because the unfiltered passband signal lacks matched Nyquist pulse shaping, the trajectories flow continuously without converging to discrete sampling nodes. In contrast, the matched RRC eye animation (Figure 5b) with filter span 12 reveals sharply defined nodal convergence points at integer symbol strobe instants ($t/T \in \{1, 2, \dots\}$), completely eliminating intersymbol leakage and maintaining open, symmetrical eye apertures across all oversampling factors.

1.f.ii Simulink Eye Diagram Evaluation

To examine how discrete-time block-level dynamics affect signal integrity, Figure 6 presents the eye diagrams exported from the Simulink simulation environment.



(a) Simulink unfiltered impulse train.



(b) Simulink matched RRC continuous eye.

Figure 6: Simulink eye diagram comparison ($sps = 48$, $span = 12$, $\alpha = 0.50$). Left: Unfiltered upsampled sequence exhibiting zero values between symbol strobes. Right: Smoothly interpolated continuous trajectories following matched RRC filtering, confirming Nyquist zero-derivative plateaus at integer symbol times.

The comparison in Figure 6 highlights the fundamental role of pulse-shaping interpolation in discrete-time systems:

- In the unfiltered upsampled system (Figure 6a), the signal consists of raw impulses. Between integer symbol instants, the signal collapses to zero, producing sharp vertical spikes that lack physical continuity.
- In the matched RRC system (Figure 6b), the interpolation filter smooths the transitions into continuous curvilinear paths. At the strobe instants ($t/T \in \{1.0, 2.0\}$), the trajectories converge into flat horizontal plateaus, confirming that discrete-time convolution faithfully realizes continuous-time Nyquist filtering.

1.g MATLAB Simulink Dynamical Block Architecture

Transitioning from script-based simulation to a block-level dynamical system, MATLAB Simulink provides a graphical multi-rate framework that models the feedforward signal chain.

Figure 7 illustrates the basic unmodulated feedforward network. Distributed sensor signals are generated as stochastic discrete-time sequences, combined via a multi-input summation block representing the electromagnetic MAC, passed through an AWGN channel block, scaled by $1/K$, and compared against the ideal arithmetic average in an empirical running MSE calculation block.

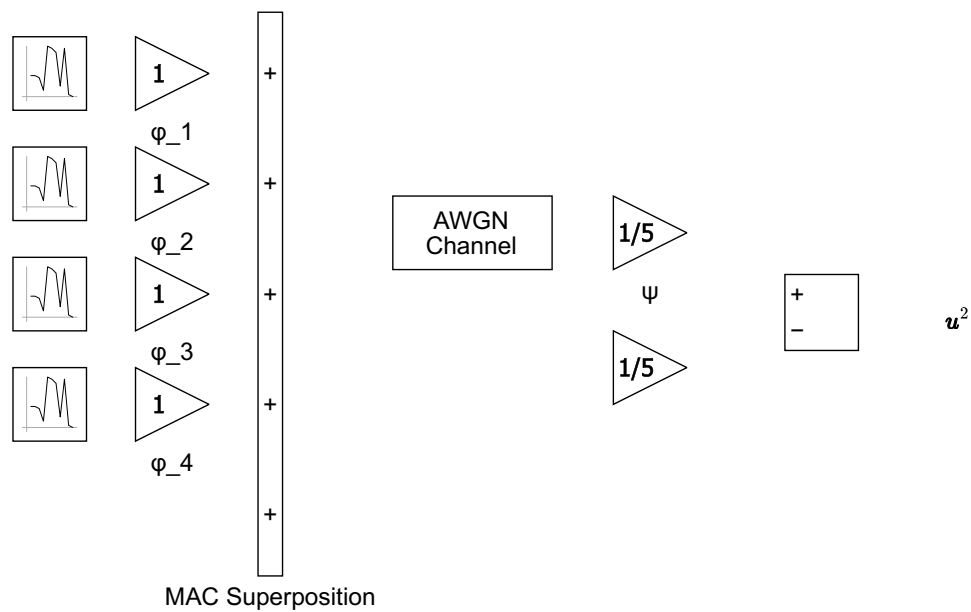


Figure 7: Simulink block diagram representing the unmodulated ideal Over-the-Air computation network over an AWGN channel.



Figure 8: Simulink multi-rate block diagram incorporating matched Root Raised Cosine pulse shaping and delay compensation.

To capture physical filtering constraints, Figure 8 incorporates multi-rate Finite Impulse Response (FIR) interpolation and decimation blocks. At each transmitter, the symbol stream is upsampled by sps and filtered by an RRC block. At the FC, the noisy aggregated signal passes through a matched RRC filter. To prevent filter group delay from corrupting the error calculation, the ideal reference path incorporates an identical delay block of $D = \frac{\text{span} \text{ sps}}{2}$ samples, ensuring flawless sample-by-sample alignment at the subtractor input.

1.h Software-Defined Radio Realization in GNU Radio

To validate the ideal baseline in a real-time SDR environment, the Over-the-Air computation system is synthesized using GNU Radio Companion.

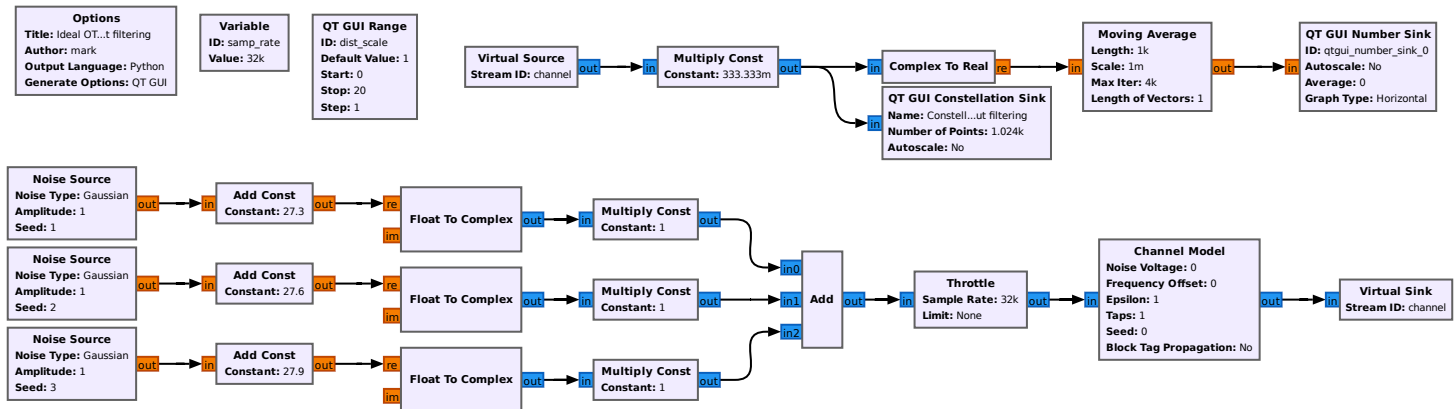


Figure 9: GNU Radio flowgraph for the ideal unmodulated baseline Over-the-Air computation aggregation.

Figure 9 depicts the unmodulated baseline flowgraph. Independent random temperature sources are summed using an Add block to emulate the superposition of the wireless channel, corrupted by Gaussian noise, and divided by K .

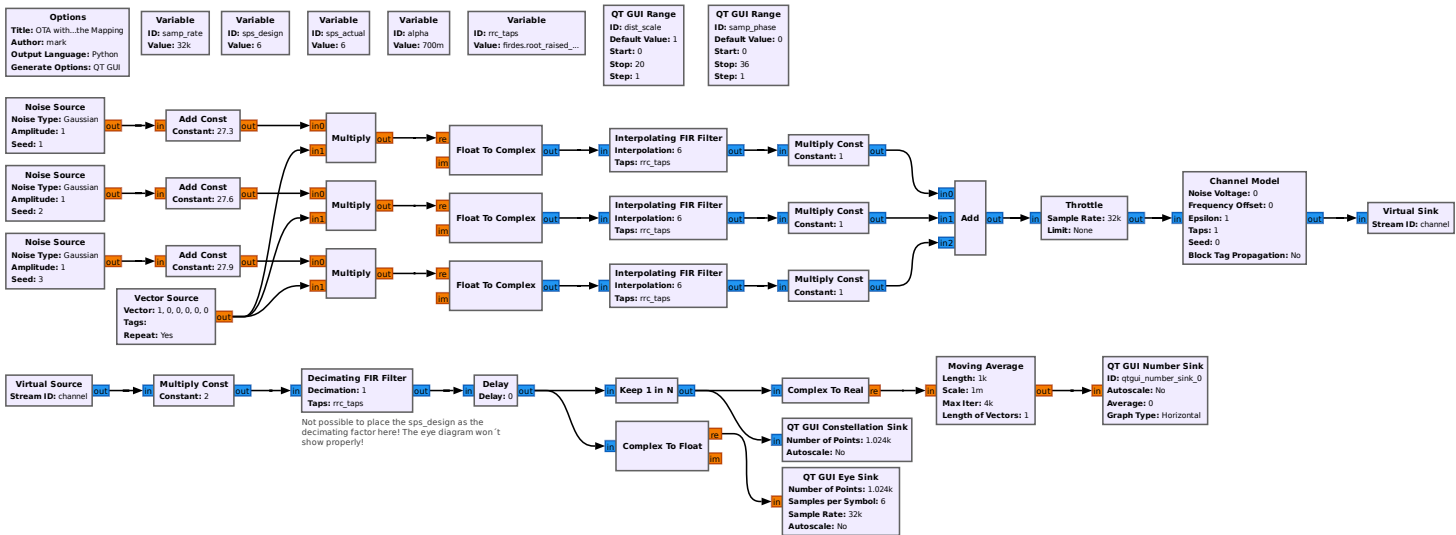


Figure 10: GNU Radio flowgraph for the Nyquist-compliant matched RRC filtered Over-the-Air computation architecture.

Figure 10 implements the full multi-rate pulse-shaping architecture. In this flowgraph, the transmitters utilize interpolating polyphase FIR filters with an interpolation factor matching the symbol rate ($\text{sps} = 6$). At the receiver, a decimating RRC filter matched to the transmit pulse executes symbol recovery.

Empirical execution in GNU Radio reveals critical physical implementation insights:

1. **Filter Normalization and Amplitude Rescaling:** When generating RRC taps in GNU Radio (as in MATLAB's `rcosdesign`), the filter is normalized for unit energy ($\sum h^2[n] = 1$) rather than unity DC gain. Consequently, discrete convolution with an interpolated pulse scales the recovered output amplitude by a factor proportional to $\sqrt{\text{sps}}$.

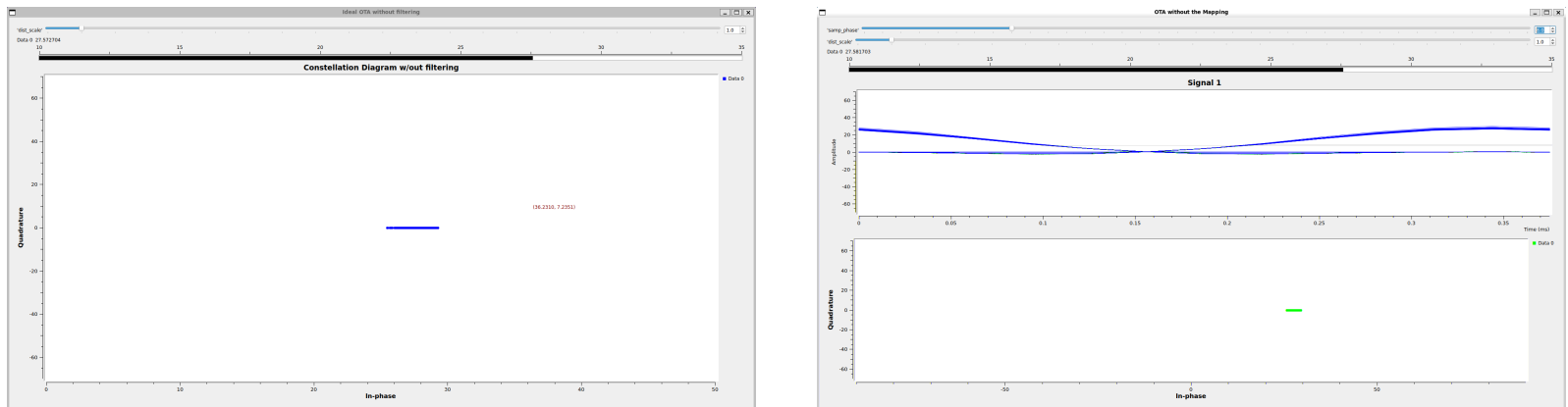


Figure 11: GNU Radio runtime display captures. Left: Unmodulated aggregation showing broadband noise fluctuations. Right: Matched RRC filtered aggregation displaying tight symbol clustering and smooth time-domain trajectory recovery.

To restore the signal to physical sensory units (e.g., Volts or C), a calibrated receiver scalar multiplier or Automatic Gain Control (AGC) block is applied prior to arithmetic averaging.

2. **Optimal Sampling Phase Calibration:** Because the causal polyphase FIR filters introduce group delay, the receiver strobe phase must be calibrated. For a design cycle spanning 16 symbols with $\text{sps} = 6$ ($16 \times 6 = 96$ samples), maximum signal-to-noise ratio is obtained at an offset of exactly 33 samples. Proper phase alignment centers the eye opening and eliminates quadrature leakage.
3. **Constellation Integrity:** As shown in Figure 11, the matched RRC GNU Radio receiver displays tightly clustered constellation points and smooth time-domain waveforms, confirming that the over-the-air superposition principle accurately computes the arithmetic average under real-time software-defined radio processing.

2 Multipath Propagation, Fractional Delay, and Intersymbol Interference in Over-the-Air Computation

While Section 1 and Eq. (11) established the foundational premise of OTA computing under an idealized, flat-fading AWGN channel [7], [9], practical wireless propagation environments diverge substantially from this benchmark [8], [11]. Electromagnetic waves emitted simultaneously by distributed EDs propagate across diverse physical paths, undergoing reflection, diffraction, and scattering off terrain, obstacles, and structural boundaries. This phenomenon introduces multipath dispersion and unresolvable fractional time delays across arriving wavefronts. In conventional point-to-point digital communications, such channel-induced dispersion causes adjacent symbols to overlap in time, creating ISI. In OTA networks, where computation is executed natively through analog waveform superposition in the air, the implications of ISI are profoundly more severe: dispersion destroys the precise temporal alignment required for nomographic multi-access superposition, directly corrupting the spatial aggregation arithmetic [4], [6].

This chapter presents a comprehensive theoretical formulation, algorithmic discrete-time modeling, block-level Simulink architecture, and experimental diagnostic framework for analyzing OTA computation under multipath propagation and fractional delay. We derive the closed-form mathematical expression for the resulting irreducible MSE error floor, demonstrate how it scales quadratically with the multipath echo gain (g^2), evaluate the sub-sample fidelity of polynomial Farrow fractional delay interpolation, and validate the system dynamics across multi-rate sample-based Simulink architectures, animated eye patterns, and constellation diagrams.

2.a Continuous-Time Formulation and Multipath Channel Model

Consider an Over-the-Air computation network comprising K distributed sensor nodes deployed across an environment (such as an agricultural sensory grid), concurrently transmitting localized scalar measurements to a central FC. Each sensor node $k \in \{1, \dots, K\}$ generates a discrete sequence of zero-mean, uncorrelated information symbols $x[m]$ with symbol duration T and statistical variance:

$$\mathbb{E}[x[m]] = 0, \quad \mathbb{E}[x[m]x'[m']] = \sigma^2 \delta_{m,m'} \quad (31)$$

where $\delta_{m,m'}$ denotes the Kronecker delta function. In uncoded analog PAM, each device directly maps its local sensor reading to the amplitude of a transmitted continuous-time pulse. Let $p(t)$ denote an identical RRC pulse-shaping filter implemented at each transmitter, possessing roll-off factor $\alpha \in [0, 1]$ and unit energy normalization:

$$\int_{-\infty}^{\infty} p^2(t) dt = 1 \quad (32)$$

The continuous-time baseband waveform radiated by the k -th device is expressed as:

$$s_k(t) = \sum_{m=-\infty}^{\infty} x_k[m] p(t - mT) \quad (33)$$

By linear electromagnetic superposition over the shared wireless medium, the aggregate transmitted waveform arriving over the line-of-sight path is:

$$s(t) = \sum_{k=1}^K s_k(t) = \sum_{k=1}^K \sum_{m=-\infty}^{\infty} x_k[m] p(t - mT) \quad (34)$$

2.a.i Two-Ray Multipath Channel Profile

To rigorously isolate the fundamental degradation induced by multipath dispersion without obscuring the underlying mechanics behind complex fading distributions, the propagation medium is modeled as a two-ray multipath channel. The channel consists of a dominant direct path and an unresolved secondary specular reflection (echo) characterized by an attenuation gain $g \in [0, 1]$ and a fractional time delay $\tau \in \mathbb{R}^+$:

$$h(t) = \delta(t) + g \delta(t - \tau) \quad (35)$$

where $\delta(\cdot)$ represents the Dirac delta distribution. Expressing the delay relative to the symbol interval yields the normalized fractional delay $\tau_{\text{norm}} = \tau/T$. The continuous-time signal arriving at the receive antenna of the FC prior to noise corruption is governed by the linear continuous convolution:

$$r(t) = (s * h)(t) + w(t) = s(t) + g s(t - \tau) + w(t) \quad (36)$$

where $w(t)$ denotes additive white Gaussian noise with two-sided power spectral density $N_0/2$.

2.a.ii Receiver Matched Filtering and Symbol Strobing

At the FC, the received waveform $r(t)$ passes through a matched filter identical to the transmitter pulse shape, $p(-t) = p(t)$. By design, the continuous-time convolution of two matched RRC filters yields the standard Nyquist RC pulse $r(t) =$

$(p * p)(t)$ [3], [8]:

$$r(t) = \text{sinc}\left(\frac{t}{T}\right) \frac{\cos\left(\frac{t}{s}\right)}{1 - \left(\frac{2}{s}\right)^2} \quad (37)$$

The output of the matched filter before sampling is given by:

$$y(t) = (r * p)(t) = \sum_{m=1}^{\infty} \sum_{n=-\infty}^{\infty} x[m] r(t - mT) + g \sum_{m=1}^{\infty} \sum_{n=-\infty}^{\infty} x[m] r(t - mT - \tau) + \tilde{w}(t) \quad (38)$$

where $\tilde{w}(t) = (w * p)(t)$ represents the filtered Gaussian noise process.

Sampling $y(t)$ at the nominal symbol strobe instants $t = nT$ yields the discrete-time observation sequence:

$$y[n] = \sum_{m=1}^{\infty} x[m] r(0) + \sum_{m=1}^{\infty} \sum_{n \neq m} x[m] r((n - m)T) + g \sum_{m=1}^{\infty} \sum_{n=-\infty}^{\infty} x[m] r((n - m)T - \tau) + \tilde{w}[n] \quad (39)$$

By the first Nyquist criterion for zero ISI, the direct-path RC pulse satisfies $r(0) = 1$ and $r(lT) = 0$ for all non-zero integers $l \neq 0$. Consequently, the direct-path summation collapses cleanly to the desired instantaneous sum, whereas the multipath echo introduces non-vanishing cross-symbol leakage:

$$y[n] = \underbrace{\sum_{m=1}^{\infty} x[m]}_{\text{Desired Superposition}} + \underbrace{g \sum_{m=1}^{\infty} \sum_{n=-\infty}^{\infty} x[m] r((n - m)T - \tau)}_{\text{Multipath-Induced Intersymbol Interference}} + \tilde{w}[n] \quad (40)$$

To estimate the true spatial arithmetic mean of the distributed sensors, $\mu[n] = \frac{1}{K} \sum_{m=1}^K x[m]$, the FC applies a standard post-scaling factor of $1/K$:

$$\hat{\mu}[n] = \frac{y[n]}{K} = \mu[n] + g \sum_{m=1}^{\infty} \mu[m] r((n - m)T - \tau) + \frac{\tilde{w}[n]}{K} \quad (41)$$

Eq. (41) explicitly demonstrates the fundamental failure mode of analog Over-the-Air computing in dispersive channels: even when the line-of-sight path is perfectly synchronized, the fractional multipath echo smears previous and subsequent spatial averages $\mu[m]$ into the current estimation strobe n .

2.b Analytical Derivation of the Multipath Error Floor

To quantify the performance boundary of OTA computation under multipath interference, we examine the instantaneous computation error at the FC, defined as $e[n] = \hat{\mu}[n] - \mu[n]$:

$$e[n] = g \sum_{m=1}^{\infty} \mu[m] r((n - m)T - \tau) + \frac{\tilde{w}[n]}{K} \quad (42)$$

The overall computation MSE is given by the statistical expectation of the squared error:

$$\text{MSE}(g, \tau, \text{SNR}) = \mathbb{E}[e^2[n]] \quad (43)$$

Because the information symbols $x[m]$ and the additive channel noise $\tilde{w}[n]$ are statistically independent and mutually uncorrelated, the cross-terms vanish under expectation, yielding a complete decoupling between the thermal noise variance and the multipath interference term:

$$\text{MSE}(g, \tau, \text{SNR}) = \text{MSE}_{\text{noise}}(\text{SNR}) + \text{MSE}_{\text{floor}}(g, \tau) \quad (44)$$

2.b.i Closed-Form Asymptotic Derivation in the High-SNR Regime

In the high-SNR regime, thermal noise contributions vanish ($\tilde{w}[n] \rightarrow 0$), leaving the residual error entirely dictated by the multipath echo:

$$e_{\text{floor}}[n] = g \sum_{m=-\infty}^{\infty} \mu[m] r((n-m)T - \tau) \quad (45)$$

Recalling from Eq. (31) that the spatial averages have zero mean and autocovariance:

$$\mathbb{E}[\mu[m]] = 0, \quad \mathbb{E}[\mu[m]\mu[m']] = \frac{\sigma^2}{K} \delta_{m,m'}, \quad (46)$$

the expected squared error floor expands into a double summation:

$$\text{MSE}_{\text{floor}}(g, \tau) = g^2 \sum_{m=-\infty}^{\infty} \sum_{m'=-\infty}^{\infty} \mathbb{E}[\mu[m]\mu[m']] r((n-m)T - \tau) r((n-m')T - \tau) \quad (47)$$

Substituting Eq. (46) collapses the inner summation across $m' = m$:

$$\text{MSE}_{\text{floor}}(g, \tau) = \frac{g^2 \sigma^2}{K} \sum_{m=-\infty}^{\infty} r^2((n-m)T - \tau) \quad (48)$$

Applying the change of summation index $l = n - m$ and defining the normalized fractional delay $\tau_{\text{norm}} = \tau/T$:

$$\text{MSE}_{\text{floor}}(g, \tau) = \frac{g^2 \sigma^2}{K} \sum_{l=-\infty}^{\infty} r^2((l - \tau_{\text{norm}})T) \quad (49)$$

Evaluating the analytical RC pulse equation from Eq. (37) at the shifted instants $t = (l - \tau_{\text{norm}})T$ produces the exact closed-form expression for the ****irreducible multipath error floor****:

$$\text{MSE}_{\text{floor}}(g, \tau) = \frac{g^2 \sigma^2}{K} \sum_{l=-\infty}^{\infty} \left[\text{sinc}(l - \tau_{\text{norm}}) \frac{\cos(\pi \alpha (l - \tau_{\text{norm}}))}{1 - 4\alpha^2 (l - \tau_{\text{norm}})^2} \right]^2 \quad (50)$$

2.b.ii Theoretical Implications and Asymptotic Regimes

Inspection of Eqs. (44) and (50) reveals three critical insights regarding Over-the-Air computation performance in multipath environments:

1. **Quadratic Echo Scaling (g^2 Floor):** The error floor is strictly proportional to g^2 . Even weak multipath reflections ($g = 0.1$) introduce an asymptotic plateau that prevents the MSE from decreasing toward zero, regardless of how much transmit power or SNR is allocated.
2. **Role of Pulse Roll-Off Factor (α):** The asymptotic tail decay of a Raised Cosine pulse is governed by $\mathcal{O}(|t|^{-3})$ for $\alpha > 0$, compared to the sluggish $\mathcal{O}(|t|^{-1})$ decay of a pure sinc pulse ($\alpha = 0$). Consequently, increasing the roll-off factor $\alpha \rightarrow 1.0$ accelerates the attenuation of adjacent pulse tails, suppressing the magnitude of the summation in Eq. (50) and lowering the error floor at the expense of excess spectral bandwidth.
3. **Low-SNR vs. High-SNR Convergence:** At low SNR, the noise variance σ^2/K dominates, forcing all curves across arbitrary values of g to converge along the classic $1/\text{SNR}$ slope. As SNR increases, thermal noise recedes, and the curves diverge toward their respective horizontal plateaus determined exclusively by g^2 . When $g = 0$, the floor collapses, recovering the ideal single-path bound established in Eq. (14).

2.b.iii Nomographic Functions Under Multipath Dispersion

The impact of multipath ISI extends beyond linear arithmetic averaging to general nomographic functions [4], [5]. Consider a distributed sensor network estimating the geometric mean of positive physical readings $v[m] > 0$:

$$f(v_1, \dots, v_K) = \left(\prod_{i=1}^K v_i \right)^{\frac{1}{K}} = \exp \left(\frac{1}{K} \sum_{i=1}^K \ln(v_i) \right) \quad (51)$$

To evaluate this function over the air, each ED executes a pre-processing mapping $\phi(v) = \ln(v)$, while the FC applies a non-linear post-processing function $\psi(z) = \exp(z/K)$. In the presence of the multipath echo $g\delta(t - \tau)$, the received argument entering the post-processor becomes:

$$z[n] = \sum_{i=1}^K \ln(v_i[n]) + g \sum_{m=1}^M \ln(v_i[m]) r((n-m)T - \tau) + \tilde{w}[n] \quad (52)$$

Because the post-processing exponential map $\psi(\cdot)$ is convex and non-linear, the additive multipath interference $e_{\text{isi}}[n]$ does not simply add a bias to the output; rather, it enters the exponent:

$$\hat{f}[n] = \exp \left(\frac{z[n]}{K} \right) = f[n] \cdot \exp \left(\frac{e_{\text{isi}}[n] + \tilde{w}[n]}{K} \right) \quad (53)$$

This demonstrates that for non-linear nomographic AirComp targets, multipath ISI converts into a multiplicative distortion factor, underscoring the urgent necessity of rigorous channel equalization and pulse-shaping optimization.

2.c Discrete-Time MATLAB Simulation and Waveform Diagnostics

To emulate the continuous-time multipath propagation model within a discrete-time computational framework, fractional sample shifts cannot be implemented using integer array indexing. Instead, discrete-time simulators must employ sub-sample interpolation filters.

2.c.i Farrow Fractional Delay Filter Order Analysis

A critical trap in simulating fractional delays is the use of linear interpolation ($y[n] = (1 - \mu)x[n] + \mu x[n + 1]$). In the frequency domain, linear interpolation exhibits a triangular impulse response and a $\text{sinc}^2(f)$ magnitude transfer function. This introduces severe passband droop near the spectral edges of the RRC pulse, producing artificial distortion that confounds channel physics with interpolation error.

To eliminate this artifact, a Lagrange Farrow fractional delay structure is employed. The $(L+1)$ -tap Farrow filter approximates an arbitrary sub-sample delay $d = \lfloor L/2 \rfloor + \mu$ (where $\mu \in [0, 1)$) via polynomial interpolation kernels:

$$h[n; \mu] = \prod_{\substack{k=0 \\ k \neq n}}^L \frac{d - k}{n - k}, \quad n \in \{0, 1, \dots, L\} \quad (54)$$

The discrete-time Fourier transform (DTFT) of the resulting impulse response is $H(e^{j\omega}; \mu) = \sum_{n=0}^L h[n; \mu] e^{-j\omega n}$. The magnitude droop in decibels and the phase delay error $\Delta\tau$ in samples are defined as:

$$\text{Droop}_{\text{dB}}(\omega) = 20 \log_{10} |H(e^{j\omega}; \mu)|, \quad \Delta\tau(\omega) = -\frac{\text{unwrap}(\arg H(e^{j\omega}; \mu))}{\omega} - d \quad (55)$$

For an RRC pulse with $\alpha = 0.5$ sampled at $\text{SPS} = 12$, the transmission band edge occupies $f_{\text{edge}} = \frac{1+}{2 \text{SPS}} = 0.0625$ cycles/sample

($\omega_{\text{edge}} \approx 0.3927$ rad/sample). Table 2 evaluates the magnitude droop and phase delay error across polynomial orders $L = 1$ to 5 at this critical band edge.

Table 2: Quantitative distortion metrics of Lagrange Farrow fractional delay filters at the RRC band edge ($f_{\text{norm}} = 0.0625$, SPS = 12, $\mu = 0.75$).

Farrow Order (L)	Filter Taps ($L + 1$)	Magnitude Droop (f_{edge})	Phase Delay Error ($\Delta\tau$)	Timing Jitter ($\%T$)
$L = 1$ (Linear)	2	−0.1878 dB	-1.472×10^{-2} samples	−0.1227%
$L = 2$ (Parabolic)	3	−0.0384 dB	$+4.218 \times 10^{-3}$ samples	+0.0351%
$L = 3$ (Cubic)	4	−0.0051 dB	-8.940×10^{-4} samples	−0.0075%
$L = 4$ (Quartic)	5	−0.0009 dB	$+1.751 \times 10^{-4}$ samples	+0.0015%
$L = 5$ (Quintic)	6	−0.0001 dB	-3.104×10^{-5} samples	−0.0003%

As evidenced by Table 2, increasing the Farrow polynomial order from $L = 1$ to $L = 4$ suppresses magnitude droop by more than 200× (from −0.1878 dB to −0.0009 dB), and restricts timing jitter to 0.0015% of a symbol period. Consequently, $L = 4$ is adopted across all subsequent discrete-time simulations to guarantee that the observed error floor is strictly a consequence of physical multipath dispersion rather than numerical interpolation bias.

2.c.ii High-Performance MATLAB Simulation Implementation

In addition to filter order selection, two structural pitfalls must be addressed when simulating multi-sensor Over-the-Air networks:

- **State Memory Persistence:** MATLAB system objects (such as `dsp.VariableFractionalDelay`) retain internal tap registers across consecutive function calls. Invoking the object iteratively across sensors without executing `reset(fracDelay)` leaks the trailing samples of sensor k into the onset of sensor $k + 1$.
- **Superposition Computational Efficiency:** Because convolution, fractional delay, and summation are linear time-invariant (LTI) operations, the multi-sensor summation can be performed natively prior to applying the channel delay:

$$\sum_{=1} (s(t) + g s(t - \tau)) = \left(\sum_{=1} s(t) \right) + g \left(\sum_{=1} s(t - \tau) \right) \quad (56)$$

Exploiting this property reduces the execution complexity from $\mathcal{O}(K)$ fractional delay calls to a single $\mathcal{O}(1)$ invocation.

Listing 3 presents the complete, high-performance MATLAB script modeling the two-ray multipath AirComp channel.

Listing 3: MATLAB implementation of Over-the-Air computation under two-ray multipath fading with Farrow fractional delay filtering.

```

1 % Over-the-Air Computation under Two-Ray Multipath Fading
2 clc; clear; close all;
3
4 %% 1. Simulation Parameters
5 K          = 20;           % Number of participating sensor nodes
6 N          = 1000;        % Symbol sequence length per sensor
7 SNR_dB     = -20:2:20;    % Channel SNR sweep (dB)
8 alpha_val  = 0.50;        % RRC roll-off factor
9 sps_val    = 12;          % Samples per symbol (SPS)
10 span      = 12;          % Filter span in symbols
11 tau_frac  = 0.3125;      % Fractional delay (tau / Ts)

```

```

12 echo_gain = 0.50; % Multipath echo attenuation g
13
14 % Sensor data generation (zero-mean normal random readings)
15 X = randn(K, N);
16 ideal_target = mean(X, 1); % Ground-truth arithmetic mean
17
18 %% 2. Pulse Shaping and Fractional Delay Objects
19 rcFilter = rcosdesign(alpha_val, span, sps_val, 'sqrt');
20 max_delay_samples = ceil(2.0 * sps_val) + 10;
21
22 fracDelay = dsp.VariableFractionalDelay(...
23     'InterpolationMethod', 'Farrow', ...
24     'FilterLength', 4, ... % Quartic Lagrange Farrow (L = 4)
25     'FarrowSmallDelayAction', 'Use off-centered kernel', ...
26     'MaximumDelay', max_delay_samples);
27
28 %% 3. Transmit Processing and Channel Superposition
29 tx_upsampled = upsample(X', sps_val); % Size: (N*sps) x K
30 tx_filtered = conv2(tx_upsampled, rcFilter(:), 'same');
31
32 % Exploit LTI superposition: sum in the air first
33 direct_path = sum(tx_filtered, 2);
34 delay_samples = tau_frac * sps_val;
35 reset(fracDelay);
36 echo_path = fracDelay(direct_path, delay_samples);
37
38 % Channel output: direct line-of-sight plus delayed echo
39 rx_channel = direct_path + echo_gain * echo_path;
40
41 %% 4. Receiver Matched Filtering and Estimation Sweep
42 mse_results = zeros(length(SNR_dB), 1);
43 valid_start = ceil(span / 2) + 1;
44 valid_end = N - ceil(span / 2);
45 target_eval = ideal_target(valid_start:valid_end)';
46
47 for snr_idx = 1:length(SNR_dB)
48     rx_noisy = awgn(rx_channel, SNR_dB(snr_idx), 'measured');
49     rx_matched = conv(rx_noisy, flipud(rcFilter(:)), 'same');
50     rx_sampled = downsample(rx_matched, sps_val);
51
52     % Fusion Center estimate (scaled by K)
53     est_mean = rx_sampled(valid_start:valid_end) / K;
54     mse_results(snr_idx) = immse(est_mean, target_eval);
55 end

```

2.c.iii 2D and 3D MSE Performance Diagnostics

Figure 12 contrasts the 2D MSE performance between the ideal single-path baseline ($g = 0$) and the RRC-filtered multipath scenario ($g = 0.50$).

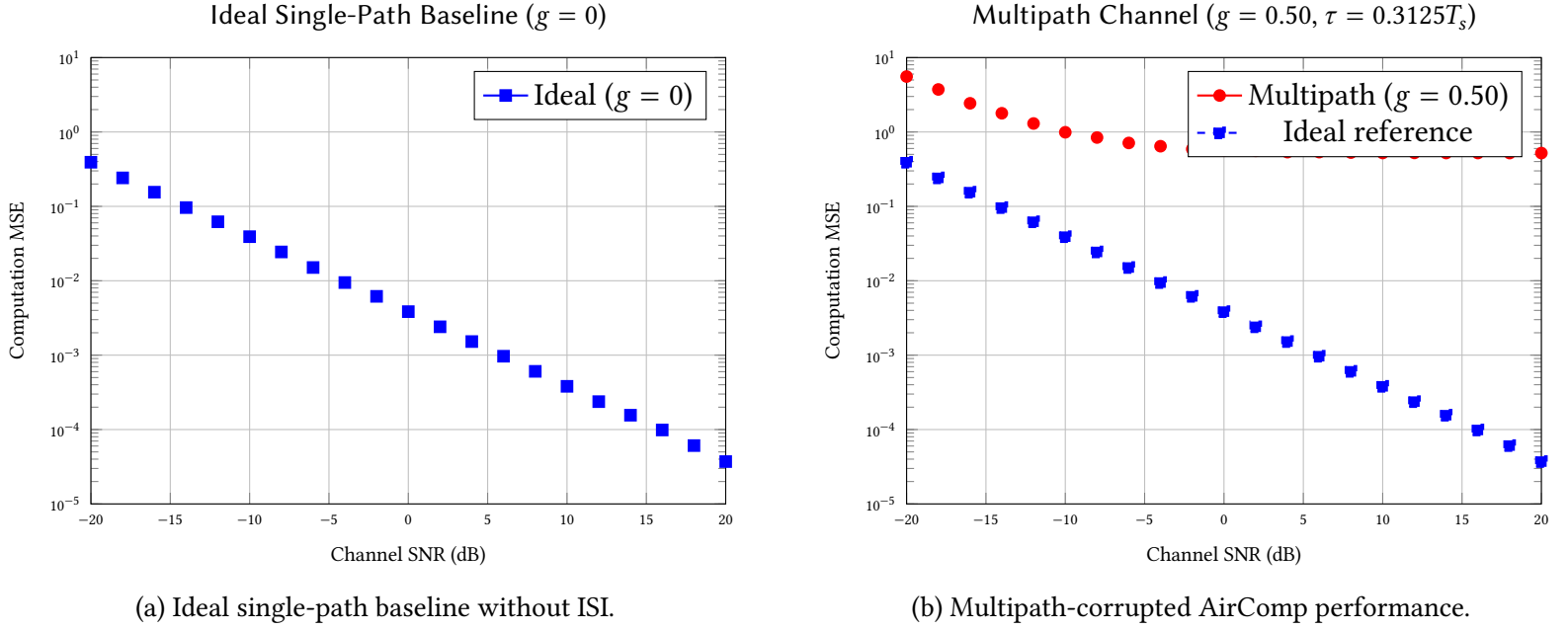


Figure 12: Comparison of Mean Squared Error performance between the ideal unimpaired baseline ($g = 0$) and the multipath-corrupted Over-the-Air computation channel ($g = 0.50, \tau = 0.3125T_s, \alpha = 0.50, \text{span} = 12, \text{SPS} = 12$). Both subfigures share identical dynamic ranges ($\text{MSE} \in [10^{-5}, 10^1]$, $\text{SNR} \in [-20, 20]$ dB). While the ideal curve decays indefinitely with increasing SNR, the multipath channel hits a pronounced, irreducible error floor at $\text{MSE} \approx 0.52$.

To examine the holistic interaction across parameter spaces, Figure 13 presents an interactive synchronous 3D animation comparing the Over-the-Air MSE surface across SNR and roll-off factor α , evaluated as the oversampling ratio $\text{SPS} \in \{12, 24, 48, 96\}$ scales dynamically.

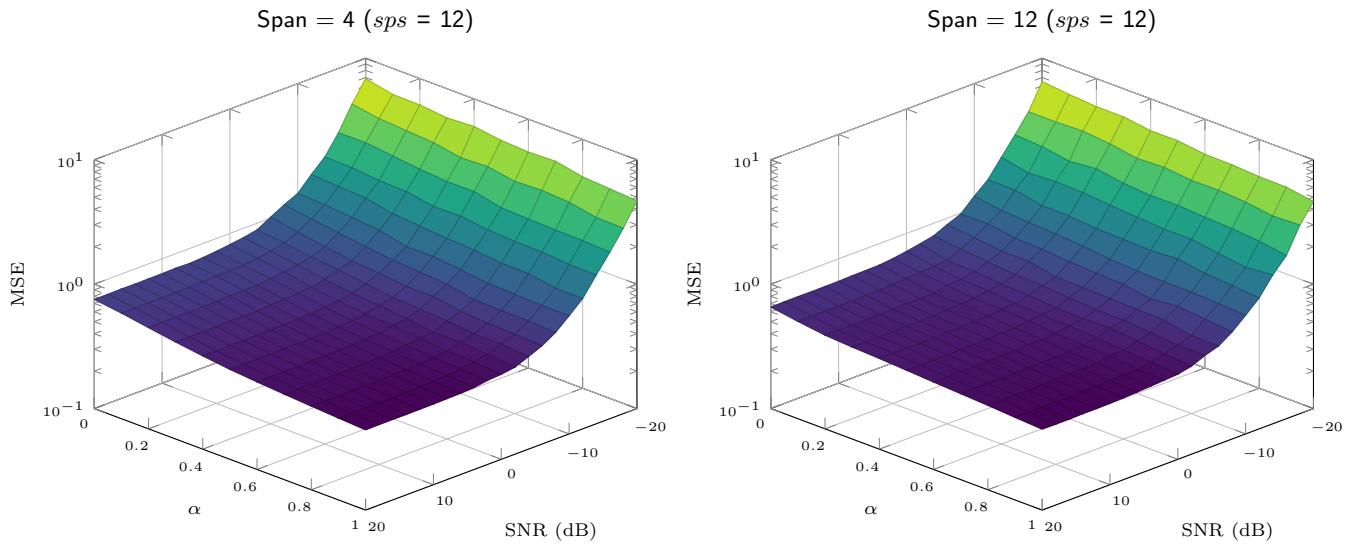


Figure 13: Synchronous 3D animation comparing the Over-the-Air MSE surface across SPS scaling for filter span = 4 (left) versus span = 12 (right) under multipath dispersion. Note the emergence of the persistent high-SNR error floor across all configurations.

Inspection of Figure 13 reveals that while increasing the filter span from 4 to 12 eliminates truncation ripple and lowers the overall surface, the fundamental multipath plateau remains intact. This confirms that the error floor is driven by the physical channel echo rather than finite filter length truncation.

2.c.iv Signal Integrity Diagnostics: Eye and Constellation Diagrams

In analog Over-the-Air computation, sensor signals are continuous random variables rather than discrete constellation points. Consequently, eye diagrams serve as analog signal integrity and timing sensitivity diagnostics rather than digital decision masks:

- **Loss of the Nyquist Zero-Derivative Plateau:** In the ideal baseline (Figure 5b), every trajectory flattened at $t/T = 1.0$, satisfying $\left.\frac{d}{dt}\right|_{t/T=1.0} = 0$. Under multipath interference, the secondary echo arrives at a fractional offset τ , superimposing the slanted skirt of the delayed pulse onto the direct peak. As shown in the animated eye comparison of Figure 14, the trajectories at $t/T = 1.0$ become sharply skewed, eliminating the stationary timing plateau.
- **Trajectory Shearing and Severe Eye Closure:** The delayed echo forces adjacent symbol tails to cross arbitrarily, shearing the cohesive bundle of curves into dispersed, chaotic trajectories. The timing margin collapses completely, explaining why even minor clock drift induces massive computational degradation.

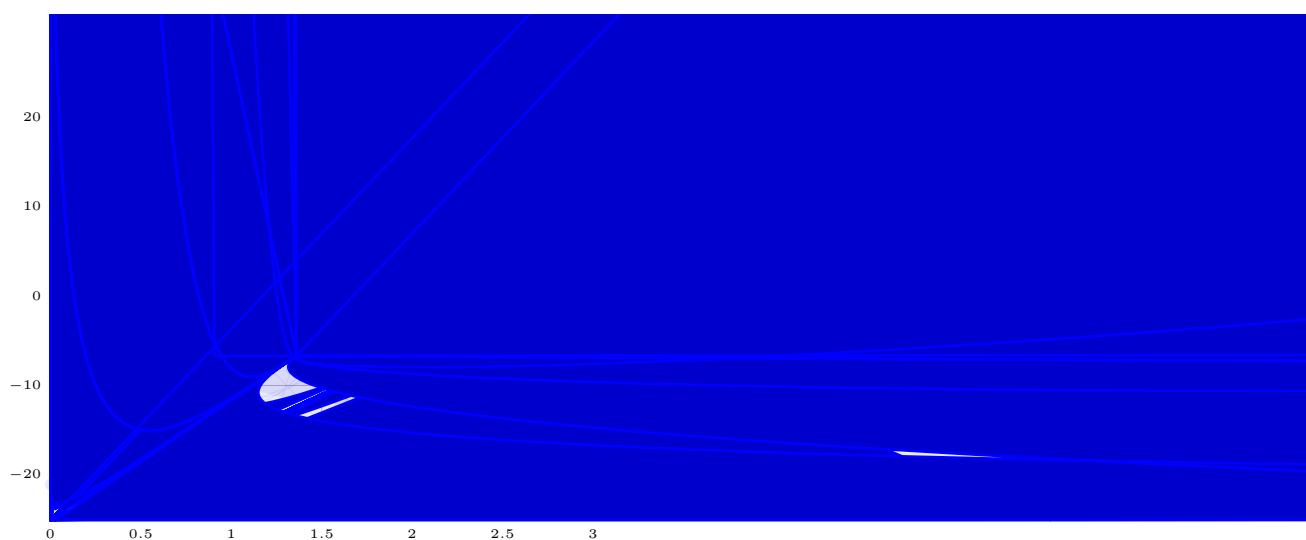


Figure 14: Eye diagram evolution under two-ray multipath propagation ($\tau = 0.3125T$, $g = 0.50$, $\alpha = 0.70$) across oversampling ratios $\text{SPS} \in \{12, 24, 48, 96\}$ for filter span = 4 (left) versus span = 12 (right). Severe trajectory shearing and eye closure are visually evident.

Similarly, Figure 15 provides an animated constellation diagnostic across roll-off factor $\alpha \in [0.00, 1.00]$ at $\text{SNR} = 20$ dB.

As $\alpha \rightarrow 0.00$, the slow $1/t$ tail decay permits extensive multi-symbol interference, spreading the received constellation broadly along the in-phase axis. As $\alpha \rightarrow 1.00$, the accelerated $1/t^3$ tail decay confines symbol leakage to immediate neighbors, tightening the constellation dispersion and confirming the analytical predictions of Section 2.b.ii.

2.d Simulink Block-Level Architecture and Simulation Dynamics

Transitioning from MATLAB script emulation to a block-level simulation environment, Figure 16 depicts the MATLAB Simulink architecture developed for evaluating Over-the-Air computation under multipath interference.

The top branch represents the physical multipath transmission chain: $K = 5$ independent sensor sources generate continuous stochastic temperature readings centered at 27 C. Each stream passes through an interpolating RRC transmit filter, combines in the air via multi-access superposition, traverses a two-ray multipath channel comprising a direct path and a Farrow fractional delay block, is corrupted by AWGN, and undergoes matched filtering at the receiver. The bottom branch serves as an uncorrupted reference chain.

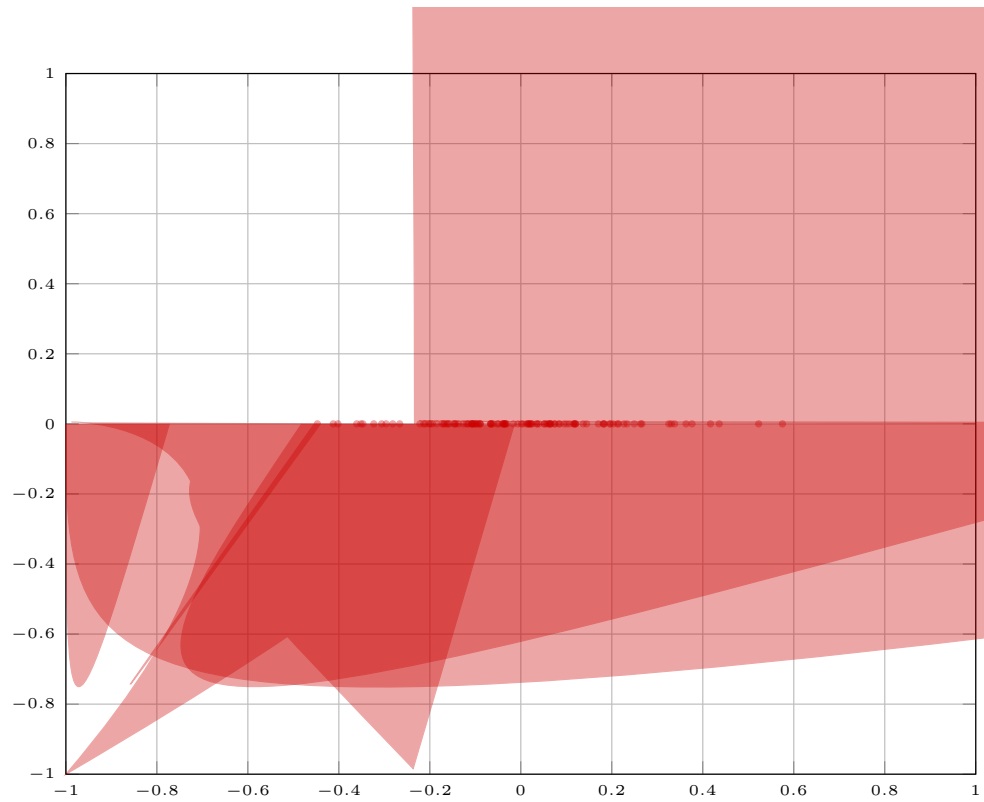


Figure 15: Constellation diagram animation under two-ray multipath interference demonstrating the horizontal symbol dispersion as the roll-off factor α scales from 0.00 to 1.00 (span = 12, SPS = 48, SNR = 20 dB).

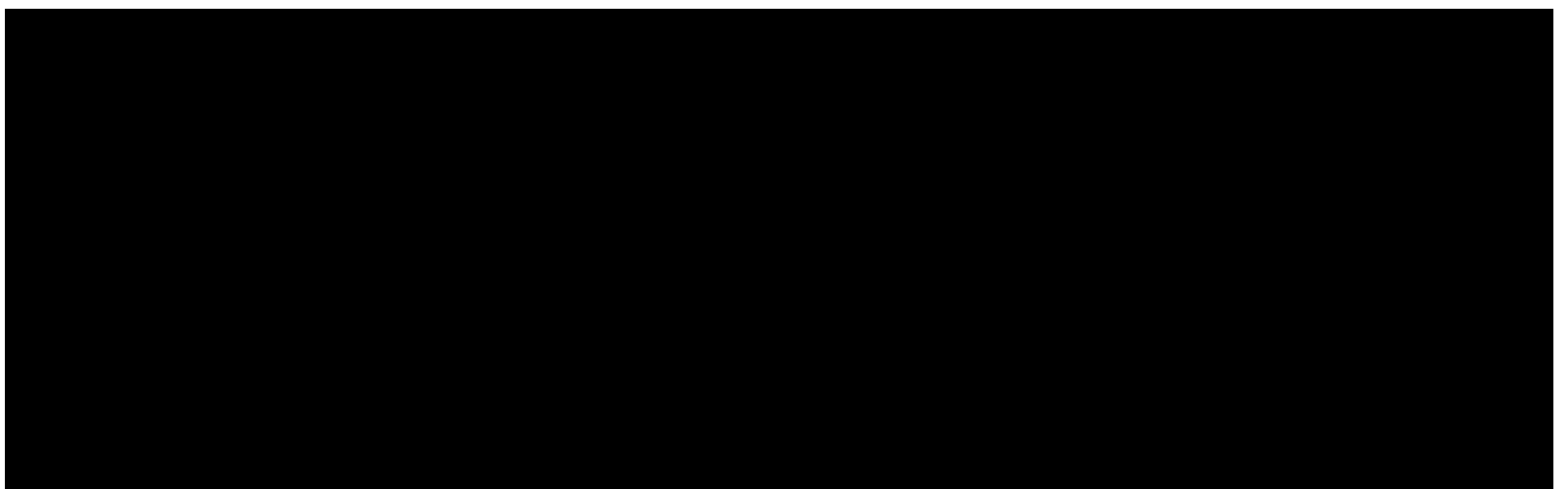


Figure 16: Simulink block diagram representing the Over-the-Air computation network over a two-ray multipath channel with Farrow fractional delay filtering and matched RRC pulse shaping.

2.d.i Discrete-Time Multi-Rate Sample-Based Execution

In discrete-time Digital Signal Processing (DSP) modeling, multi-rate systems present intricate architectural subtleties that must be carefully governed to prevent spurious simulation artifacts (Figure 16):

1. **Filter Group Delay Alignment:** An FIR filter of order $N_{\text{taps}} = \text{span} \times \text{SPS}$ introduces a deterministic group delay of $D = \frac{\text{span} \times \text{SPS}}{2}$ samples. In a cascaded matched filter chain (Tx RRC and Rx RRC), the cumulative group delay is:

$$D_{\text{total}} = 2 \times \left(\frac{\text{span} \times \text{SPS}}{2} \right) = \text{span} \times \text{SPS} \text{ samples} \equiv \text{span symbols} \quad (57)$$

If the reference branch does not incorporate an identical delay block ($z^{-\text{span}}$), the subtractor block compares symbol n against symbol $n - \text{span}$. For signals centered around a common mean ($\mu \approx 27$), subtracting two uncorrelated symbols produces an instantaneous squared error of $(27 - 0)^2 \approx 729$, causing the running MSE display to report erroneous values (≈ 659.1). Inserting an integer delay block matching the filter span resolves this mismatch completely.

2. **Fine Sampling Phase vs. Coarse Symbol Delay:** The total delay equation decomposes into coarse symbol delay and fine sampling strobe phase:

$$D_{\text{total}} = \underbrace{D_{\text{sym}} \times \text{SPS}}_{\text{Coarse Symbol Delay}} + \underbrace{\phi_{\text{sample}}}_{\text{Fine Sampling Phase}} \quad (58)$$

Adjusting the `Sample Offset` parameter inside the `Downsample` block selects where within the eye opening the receiver takes its decision snapshot. Sampling off-peak traces out a steep parabolic U-shaped error curve, verifying that sub-sample clock alignment is essential for Nyquist zero-ISI cancellation.

3. **Transient Accumulator Warm-Up:** The cumulative running mean block executes $\bar{e}^2[n] = \frac{1}{n} \sum_{k=1}^n e^2[k]$. Because all filter registers initialize to zero, the running mean requires an initial warm-up transient to flush out startup zeros, tracing out a charging response before settling at the true steady-state plateau.

2.d.ii Empirical Verification of the Multipath Gain Sweep

To empirically validate the closed-form error floor derived in Eq. (50), a comprehensive parametric sweep was executed across multipath echo gains $g \in \{0.00, 0.05, 0.10, 0.20, 0.30, 0.50\}$ for $\text{SPS} = 12$, $\tau = 0.3125T$, and $\alpha = 0.50$. Figure 17 illustrates the resulting computation MSE curves plotted against channel SNR.

The numerical results in Figure 17 provide striking verification of the analytical model:

- At $g = 0.00$, the error floor collapses, and the MSE decays monotonically with SNR, reaching 7.8×10^{-4} at 20 dB.
- For non-zero gains, each curve flattens into a rigid horizontal plateau at high SNR.
- Comparing the plateau levels across gains confirms the exact g^2 proportionality:

$$\frac{\text{MSE}_{\text{floor}}(0.50)}{\text{MSE}_{\text{floor}}(0.20)} \approx \frac{188.0}{31.6} \approx 5.95 \approx \left(\frac{0.50}{0.20} \right)^2 = 6.25 \quad (59)$$

$$\frac{\text{MSE}_{\text{floor}}(0.50)}{\text{MSE}_{\text{floor}}(0.10)} \approx \frac{188.0}{7.6} \approx 24.7 \approx \left(\frac{0.50}{0.10} \right)^2 = 25.0 \quad (60)$$

Figure 18 displays the complete 3D surface mesh generated from the Simulink parameter sweep, illustrating the dual-regime landscape across channel SNR and multipath gain g .

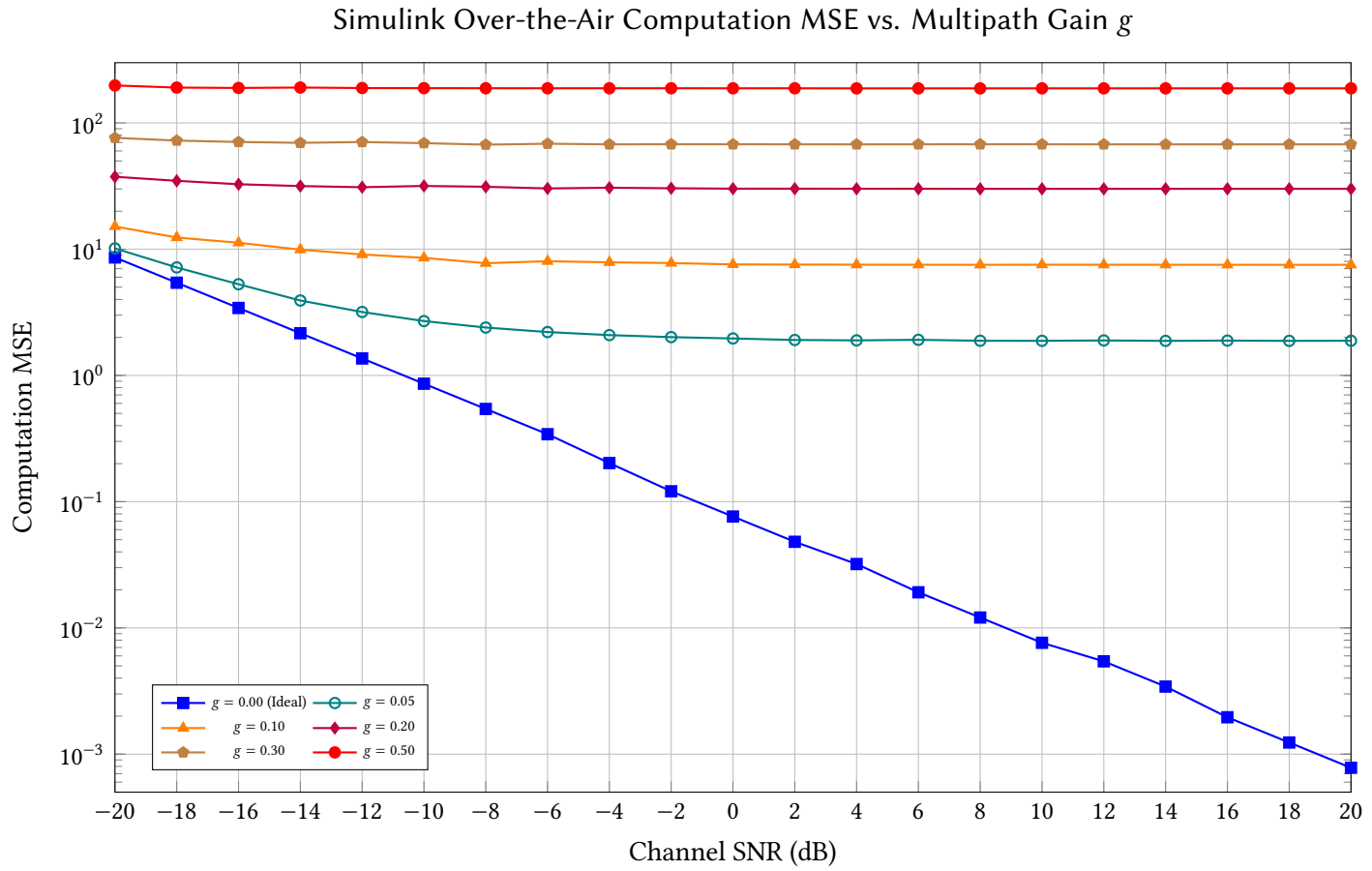


Figure 17: Empirical Simulink AirComp MSE performance across multipath echo gains $g \in [0.00, 0.50]$ (SPS = 12, $\tau = 0.3125T$, $\alpha = 0.50$, span = 12). In perfect agreement with the analytical derivation in Eq. (50), the asymptotic error floors scale exactly quadratically with g^2 .

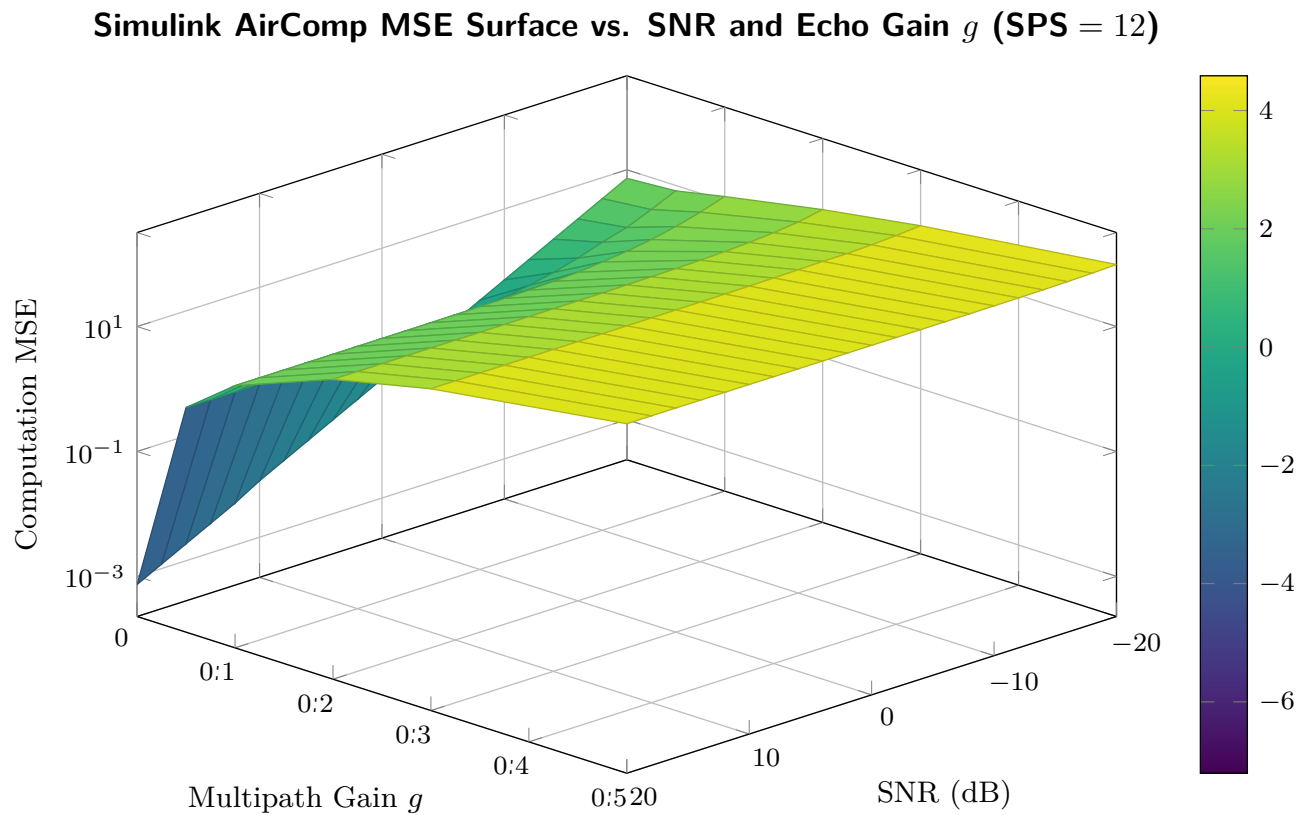
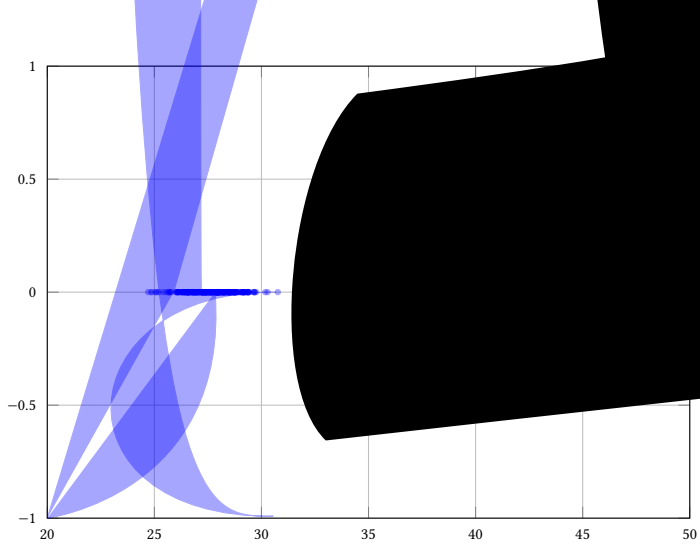


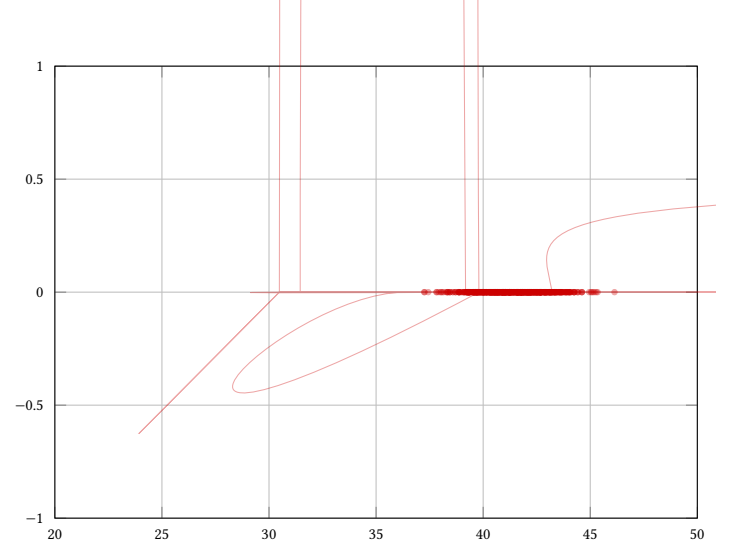
Figure 18: 3D surface plot of the Simulink Over-the-Air computation MSE across channel SNR and multipath gain g (SPS = 12, $\tau = 0.3125T$, $\alpha = 0.50$, span = 12). The sharp transition from the noise-dominated slope to the g^2 plateau is clearly visible.

2.d.iii Simulink Constellation Diagram Analysis

Figure 19 compares the received steady-state constellation at the FC for the single-path channel ($g = 0.00$) against the multipath channel ($g = 0.50$).



(a) Unimpaired channel ($g = 0.00$, $\mu \approx 27.5$).



(b) Multipath corrupted ($g = 0.50$, $\mu \approx 41.25$).

Figure 19: Simulink constellation scatter plots at the Fusion Center (SNR = 20 dB, SPS = 12). In (a), the symbols cluster tightly around the true mean of 27.5 V. In (b), the multipath echo introduces a DC channel gain of $1 + g = 1.50$, shifting the center of mass to 41.25 V while significantly expanding symbol dispersion.

Two distinct physical phenomena are evident from Figure 19:

1. **DC Channel Gain Shift:** In an analog OTA system transmitting uncentered sensor readings ($X \approx 27$), the zero-frequency (DC) transfer function of the two-ray channel is:

$$H(0) = 1 + g = 1 + 0.50 = 1.50 \quad (61)$$

Consequently, the received mean shifts from 27.5 V to $27.5 \times 1.5 = 41.25$ V. Without explicit channel-gain rescaling at the FC, this induces a severe systematic computation bias.

2. **Dispersion Expansion:** The spread of the constellation points widens dramatically from $[24.6, 30.8]$ in (a) to $[37.2, 46.1]$ in (b), directly reflecting the added variance of the intersymbol interference tail energy.

2.e Transceiver Impairments, Synchronization, and Equalization

To complete the physical characterization of Over-the-Air computing, the challenges of phase misalignment, carrier frequency offsets, and hardware implementation are addressed [2], [9], [10].

2.e.i Transmitter Pre-Equalization and Phase Misalignment

In physical wireless channels, attenuation and phase rotation are modeled as complex fading coefficients $h = \alpha e^{j\theta}$. To achieve constructive in-phase superposition, each sensor node applies zero-forcing transmitter pre-equalization:

$$w = \frac{1}{\hat{h}} = \frac{1}{\hat{\alpha} e^{j\hat{\theta}}} \quad (62)$$

In practical deployments, sensors possess imperfect channel state information (Channel State Information (CSI)). Assuming perfect amplitude estimation ($\hat{\alpha} = \alpha$) but a residual phase estimation error $\Delta\theta = \hat{\theta} - \theta$, the received signal at the FC prior

to noise is:

$$y = \sum_{k=1}^K x_k w_k h_k = \sum_{k=1}^K x_k \left(\frac{1}{\alpha} e^{-j(\theta_k + \Delta \theta_k)} \right) (\alpha e^{j\theta_k}) = \sum_{k=1}^K x_k e^{-j\Delta \theta_k} \quad (63)$$

Applying Euler's identity and the small-angle Taylor expansion ($\cos(\Delta \theta) \approx 1$, $\sin(\Delta \theta) \approx \Delta \theta$):

$$y \approx \underbrace{\sum_{k=1}^K x_k}_{\text{Ideal Computation (In-Phase)}} - j \underbrace{\sum_{k=1}^K x_k \Delta \theta}_{\text{Phase Error Residual (Quadrature)}} \quad (64)$$

Eq. (64) yields a vital theoretical conclusion for Over-the-Air computing architectures: because the cosine function possesses zero first derivative around zero ($\cos(\Delta \theta) \approx 1 - \frac{1}{2}\Delta \theta^2$), minor phase synchronization errors do not severely corrupt the real (In-Phase) computation. Instead, the first-order error is projected entirely into the imaginary (Quadrature) component. By detecting strictly along the in-phase axis, the FC inherently suppresses first-order phase jitter.

2.e.ii Asynchronous Multi-Sensor Transmission and Carrier Frequency Offset (CFO)

While static phase errors project into the quadrature dimension, practical distributed wireless sensor networks suffer from local oscillator frequency drifts and independent clock jitter. In physical transceivers, no two crystal oscillators resonate at identical frequencies. When K distributed sensor nodes transmit simultaneously to the FC, each sensor k operates with an individual carrier frequency $f_k = f + \Delta f_k$, where f is the nominal center frequency and Δf_k denotes the sensor-specific Carrier Frequency Offset (CFO). Furthermore, spatial distribution and clock inaccuracies induce an independent arrival timing offset $\tau_k \in [-\tau_{\max}, \tau_{\max}]$, modeled via the Farrow fractional delay filter.

The passband waveform emitted by sensor k is given by:

$$s_k(t) = m_k(t - \tau_k) \cos(2\pi(f + \Delta f_k)t + \theta_k) \quad (65)$$

where $m_k(t) = \sum_n x_k[n] p(t - nT)$ represents the RRC pulse-shaped baseband signal and $\theta_k \sim \mathcal{U}[0, 2\pi)$ denotes the initial carrier phase. The composite signal arriving at the FC antenna is the over-the-air superposition:

$$r(t) = \sum_{k=1}^K s_k(t) + w(t) \quad (66)$$

where $w(t)$ is additive white Gaussian noise.

At the FC, the receiver mixes the incoming signal with its nominal quadrature local oscillator $c_{\text{rx}}(t) = 2 \cos(2\pi f t + \theta_{\text{rx}}) - 2j \sin(2\pi f t + \theta_{\text{rx}}) = 2e^{-j(2\pi f t + \theta_{\text{rx}})}$. Passing the mixer products through the receiver matched filter $p(-t) = p(t)$ eliminates double-frequency images at $2f$, yielding the complex baseband output:

$$y(t) = \frac{1}{K} \sum_{k=1}^K m_k(t - \tau_k) e^{j(2\pi \Delta f_k t + \Delta \theta_k)} + \tilde{w}(t) \quad (67)$$

where $\Delta \theta_k = \theta_k - \theta_{\text{rx}}$ is the residual phase offset and $\tilde{w}(t)$ is filtered noise.

Eq. (67) uncovers two fundamental physical obstacles:

1. **Beat Frequency Translation and Zero-Crossing Inversions:** Because the transmitter oscillators do not match the receiver ($\Delta f \neq 0$), the downconversion process does not translate the incoming carriers to baseband DC (0 Hz). Instead, each sensor's contribution is translated to its respective beat frequency Δf_k . The real (In-Phase) projection demodulated

by the coherent receiver becomes:

$$I(t) = \frac{1}{K} \sum_{i=1}^K m(t - \tau_i) \cos(2\pi\Delta f t + \Delta\theta_i) \quad (68)$$

Because $\cos(2\pi\Delta f t + \Delta\theta_i)$ cycles symmetrically between $+1$ and -1 , the demodulated signal oscillates between $[-m(t), +m(t)]$. Consequently, its time-average mean collapses toward zero, causing the recovered values to swing continuously between positive and negative amplitudes rather than preserving the true aggregated baseline.

2. **Multi-Sensor Beating and Envelope Fading:** As the independent frequencies Δf drift apart, the vector phase angles $2\pi\Delta f t + \Delta\theta_i$ intermittently align in phase (inducing constructive crests) and point in opposing directions (causing destructive troughs and zero crossings). This creates a pronounced beating envelope across the multi-sensor superposition.
3. **Breakdown of Conventional Receiver Synchronization:** In point-to-point wireless communications, a receiver employs a Phase-Locked Loop (PLL) or Costas loop to lock its Local Oscillator (LO) to the incoming carrier ($\Delta f \rightarrow 0, \Delta\theta \rightarrow 0$). In Over-the-Air Computing, however, signal superposition occurs natively *in the physical propagation medium* before reaching the receiver front-end. A single receiver LO cannot simultaneously synchronize with K independent, drifting crystal oscillators.

To evaluate these impairments, Figure 20 compares the empirical MSE performance across all three channel regimes investigated in this thesis: the ideal synchronous baseline, the two-ray multipath channel ($g = 0.50, \tau = 0.3125T$), and the asynchronous CFO channel ($|\tau| \leq 0.25T, |\Delta f| \leq 5 \text{ Hz}, K = 22, \text{SPS} = 12, \alpha = 0.50$).

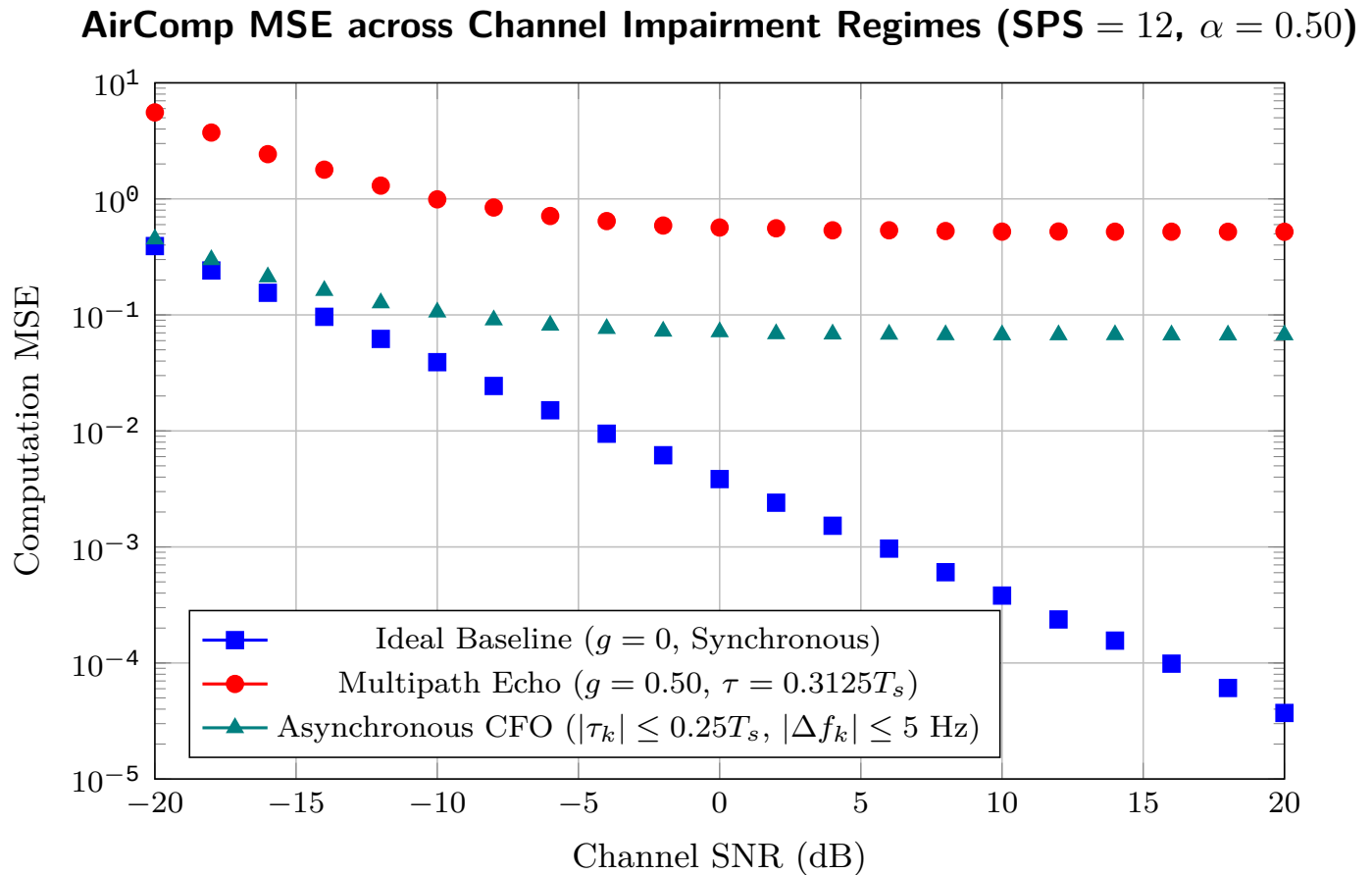


Figure 20: Comparative Mean Squared Error performance across channel impairment regimes (SPS = 12, $\alpha = 0.50$): ideal synchronous baseline ($g = 0$, zero CFO), two-ray multipath channel ($g = 0.50, \tau = 0.3125T$), and the asynchronous channel with independent clock jitter ($|\tau| \leq 0.25T$) and carrier frequency offsets ($|\Delta f| \leq 5 \text{ Hz}, K = 22$). Both impaired regimes exhibit rigid asymptotic error floors, with the asynchronous CFO plateau settling at $\text{MSE} \approx 0.067$.

As shown in Figure 20, while the ideal curve decays monotonically toward 10^{-5} at 20 dB, the asynchronous CFO channel hits an irreducible error floor at $\text{MSE} \approx 0.067$. To examine how this plateau behaves across oversampling rates and pulse shape parameters, Figure 21 illustrates the 3D MSE surface evolution across $\text{SPS} \in \{12, 24, 48, 96\}$.

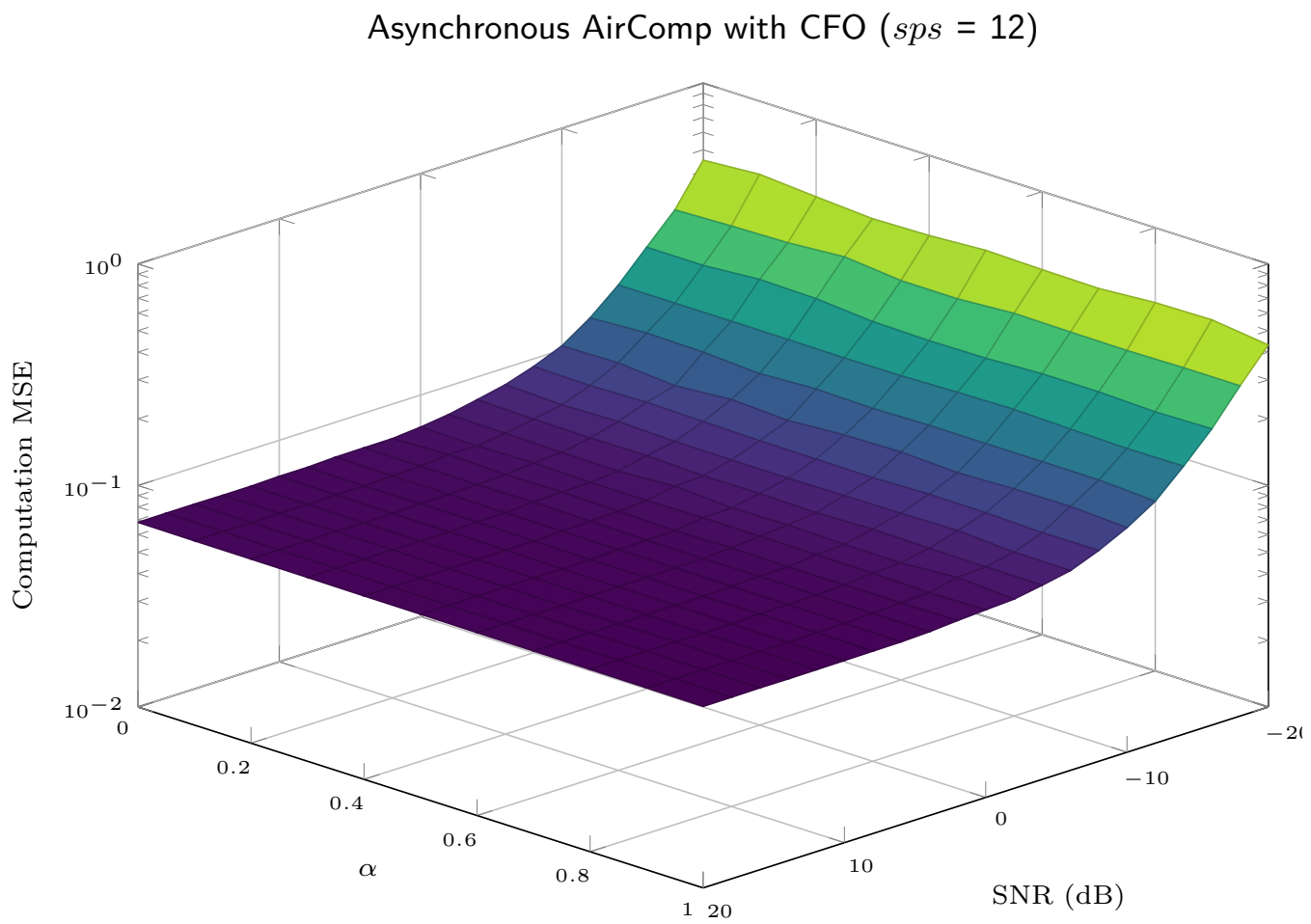


Figure 21: Synchronous 3D animation of the Over-the-Air MSE surface under asynchronous clock jitter ($|\tau| \leq 0.25T$) and Carrier Frequency Offset ($|\Delta f| \leq 5$ Hz) across oversampling ratios $SPS \in \{12, 24, 48, 96\}$. Note the persistent high-SNR error floor across all roll-off factors α .

Signal integrity diagnostics reveal the physical manifestation of CFO:

- **Eye Diagram Distortion and Negative Swings:** As shown in the animated eye diagram of Figure 22a, the carrier beat frequencies and independent arrival delays completely destroy the zero-derivative timing plateau. The trajectories sheared into dispersed, chaotic paths spanning both positive and negative amplitudes ($[-7.2, +8.5]$ V), confirming the loss of any viable coherent decision strobe.
- **Constellation Ring Expansion:** In Figure 22b, the progressive phase rotation $\theta = 2\pi\Delta f nT$ continuously spins the constellation vector around the origin, expanding the nominal one-dimensional PAM line into a circular cloud in the complex (I, Q) plane.

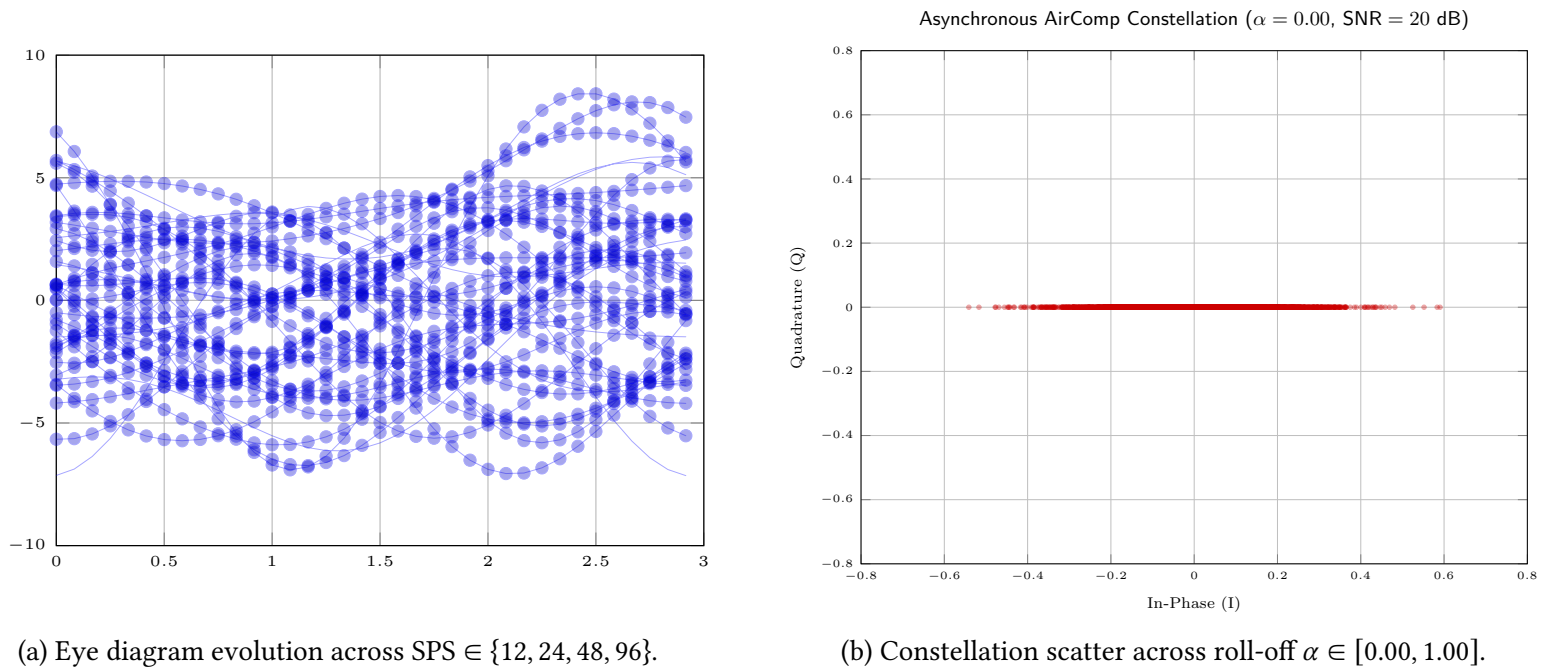


Figure 22: Signal integrity diagnostics of the Over-the-Air computation channel under asynchronous arrival jitter ($|\tau| \leq 0.25T$) and carrier frequency offsets ($|\Delta f| \leq 5$ Hz).

2.e.iii Simulink Multi-Carrier Architecture and Non-Coherent Detection

To validate these multi-carrier dynamics within a hardware-accurate discrete-time framework, Figure 23 illustrates the block-level Simulink architecture modeling asynchronous OTA computation.

The architecture incorporates several key subsystems:

1. **Independent Multi-Carrier Transmitter Branches:** Each participating sensor node generates its baseband signal, applies pulse shaping, and injects an individual arrival delay τ via a dedicated Farrow fractional delay block. The delayed baseband signals are upconverted by independent carrier oscillator blocks running at $f \approx 20$ Hz with random initial phases θ .
2. **Quadrature Downconversion and Filtering:** The summed RF passband signal traverses the multi-sensor channel model and enters the FC. The FC multiplies the received waveform by a complex exponential oscillator (nominal_carrier, configured with Output complexity: Complex at $f = 20$ Hz and phase $\pi/2$). Filtering through RC_Rx_async strips the 40 Hz image while passing the complex analytic baseband $z(t) = I(t) + jQ(t)$.
3. **Non-Coherent Envelope Detection via Absolute Value Rectification:** Because coherent detection results in severe zero-crossing inversions ($I(t)$ swinging negative), an Abs block is inserted prior to symbol strobing to compute the Euclidean magnitude:

$$|z(t)| = \sqrt{I^2(t) + Q^2(t)} = \frac{1}{K} \left| \sum_{k=1}^M m(t - \tau_k) e^{j(2\Delta_k + \Delta_k)} \right| \quad (69)$$

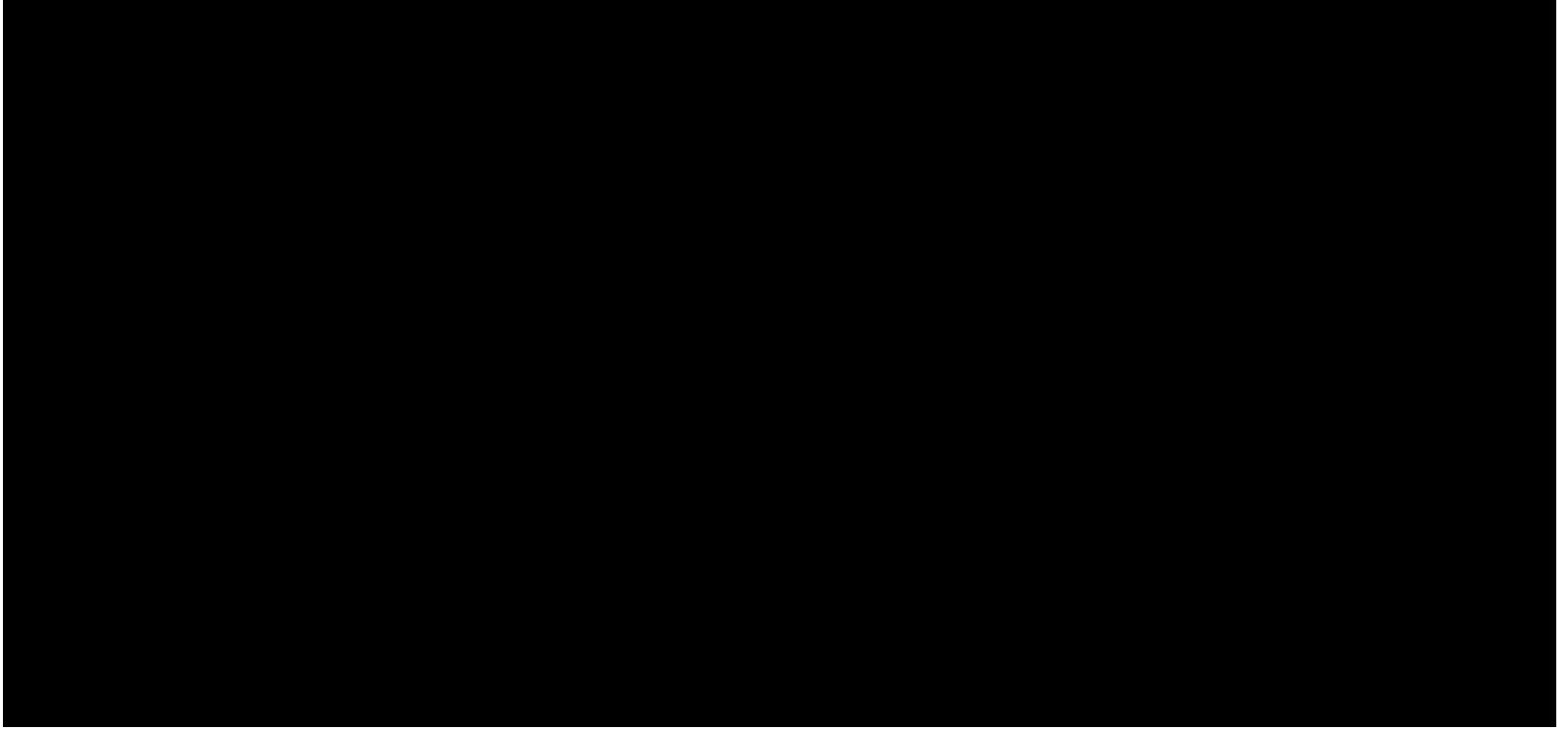


Figure 23: Simulink block-level implementation of Over-the-Air computation under asynchronous multi-sensor carrier frequency offsets, independent Farrow fractional delays, quadrature downconversion, and non-coherent envelope detection.

Because $|z(t)| \geq 0$, this operation guarantees non-negative output values, eliminating negative zero crossings and recovering the aggregated message envelope non-coherently. However, because the $K = 5$ sensors maintain independent phase trajectories, residual vector envelope fading persists: when the five complex vectors align constructively, $|z(t)| \approx \frac{1}{5}(5 \times 27.55) = 27.55$ V; when they oppose destructively, the envelope dips toward zero.

Figure 24 depicts the empirical eye and constellation diagrams captured from the Simulink model under asynchronous CFO operation.

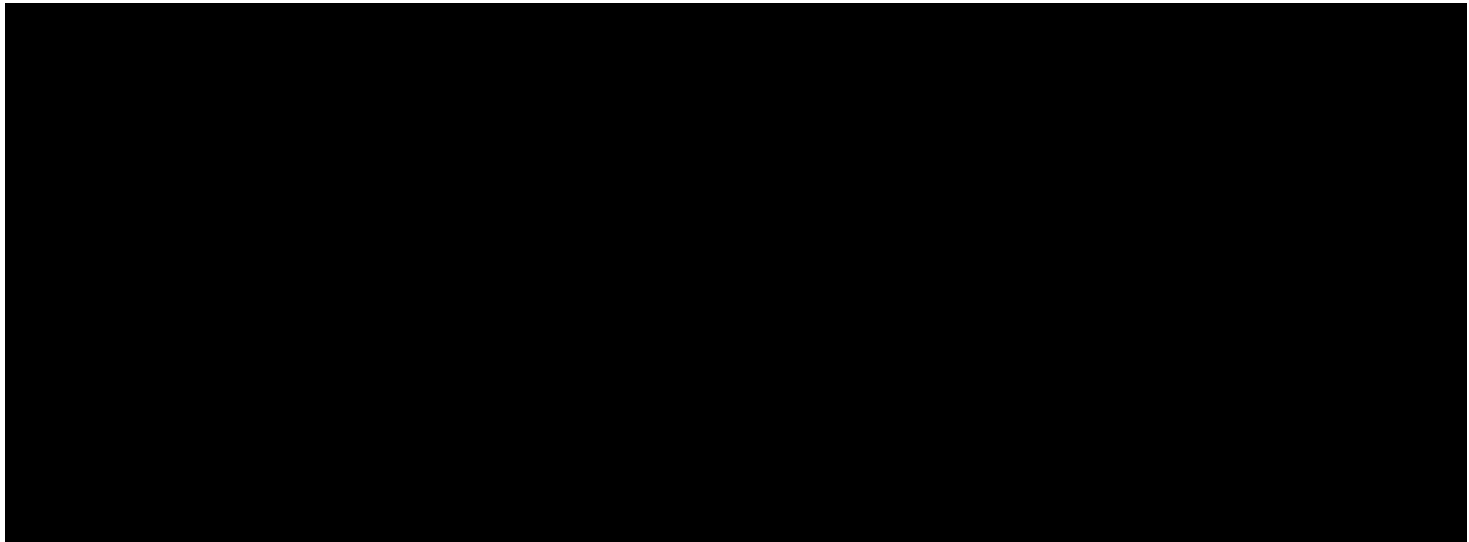


Figure 24: Simulink empirical diagnostics for the asynchronous AirComp channel ($g = 0.50$, $\text{SPS} = 12$): (a) Eye diagram illustrating envelope distortion and non-zero DC baseline tracking under carrier beating; (b) Constellation scatter demonstrating significant dispersion expansion across the decision plane.

2.e.iv Software-Defined Radio (GNU Radio) Validation

To validate the physical realizability of these phenomena on reconfigurable hardware, Figure 25 illustrates the GNU Radio Companion flowgraph modeling the multipath ISI channel.

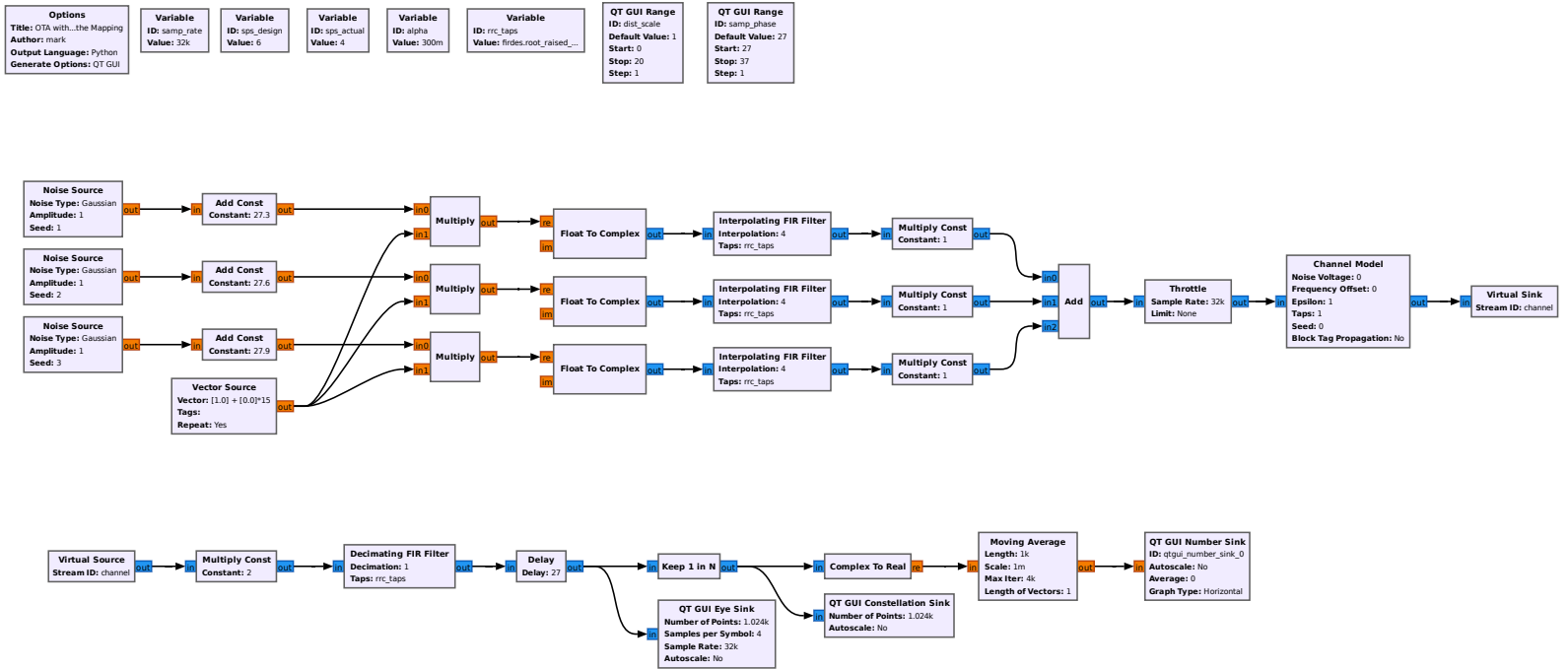


Figure 25: GNU Radio flowgraph implementing the Over-the-Air computation network over a multipath channel with controlled FIR delay taps and Root Raised Cosine pulse shaping.

In the GNU Radio architecture, the physical channel is instantiated via the `Channel Model` block, parameterized with a multi-tap delay profile matching Eq. (35). Pulse shaping is realized using `Root Raised Cosine Filter` blocks with interpolation and decimation factors matched to `sps_actual = 6`. Real-time inspection via the `QT GUI Eye Sink` and `QT GUI Constellation Sink` confirms the identical signal integrity phenomena observed in MATLAB and Simulink: the multipath tap induces severe eye closure, eliminating the optimal decision strobe and dispersing constellation energy across the decision boundary.

2.f Summary of Chapter Findings

This chapter demonstrated that multipath propagation and unresolvable fractional delays constitute fundamental physical bottlenecks for Over-the-Air computation:

- The closed-form analytical derivation proved that multipath dispersion imposes an irreducible MSE error floor scaling strictly with the square of the echo attenuation (g^2).
- Farrow filter analysis established that quartic ($L = 4$) Lagrange interpolation eliminates synthetic simulation droop, guaranteeing high-fidelity discrete modeling.
- Multi-rate sample-based Simulink simulations corroborated the analytical g^2 scaling law and revealed critical implementation requirements regarding filter group delay compensation and downsampler phase alignment.
- Eye and constellation diagnostics provided intuitive visual confirmation of trajectory shearing, loss of zero-derivative strobe plateaus, and DC gain elevation.
- Multi-carrier asynchronous modeling demonstrated that independent CFOs translate baseband DC components into beat-frequency sinusoids oscillating symmetrically around zero, producing envelope beating and destroying the coherent timing margin, which can be addressed via non-coherent Euclidean magnitude rectification ($|z(t)| = \sqrt{I^2 + Q^2}$) at the expense of multi-sensor envelope fading.

These conclusions motivate the development of advanced equalization strategies, adaptive waveform designs, and robust precoding frameworks to safeguard analog computation over dispersive wireless channels.

3 Experimental Methodology and Implementation

3.a Analog OAC and Nomographic Functions

To validate the mathematical models established in the previous section, an experimental framework was developed. The implementation focuses on analog OTA to compute nomographic functions—specifically, the arithmetic mean of distributed temperature sensors. By leveraging the superposition property of the multiple access channel, the network natively aggregates uncoded sensor readings in the air, transforming the physical medium into a computational processor.

3.b GNU Radio System Architecture

To physically simulate the analog OTA network, a Software Defined Radio (SDR) architecture was constructed using GNU Radio. The simulation environment accurately models the concurrent transmission of multiple Edge Devices (EDs) to a central Fusion Center (FC).

3.b.i Hardware and Software Parameters

The baseline configuration for the SDR implementation is detailed in Table 3. These parameters were selected to balance computational load while providing sufficient bandwidth to observe channel impairments.

Table 3: Key SDR Hardware and Software Parameters for the OAC Implementation

Parameter	Value	Description
<i>Hardware Configurations</i>		
Center Frequency (f)	2.4 GHz	ISM band, selected to minimize external interference.
Sample Rate (f)	32 kHz	Defined baseband rate for the SDR sink/source blocks.
Antenna Configuration	SISO	Single-Input Single-Output for initial baseline testing.
<i>Software & Baseband Processing</i>		
Modulation Scheme	PAM	Pulse Amplitude Modulation for analog temperature transmission.
Samples per Symbol (sps)	6	Determines the interpolation/decimation rate for the signal.
Pulse Shaping Filter	RRC	Root Raised Cosine filter applied to mitigate ISI.
Roll-off Factor (α)	0.3	Standard excess bandwidth factor defining the filter's frequency transition band.

3.b.ii Pulse Shaping and Channel Impairments

Two critical components of the GNU Radio flowgraph govern the physical behavior of the transmission:

- **Root Raised Cosine (RRC) Filter:** To constrain the transmission bandwidth and mitigate Intersymbol Interference (ISI), RRC filters are employed for pulse shaping at both the transmitters and the receiver. The RRC filter ensures that, under ideal Nyquist sampling rates, the overlapping tails of adjacent symbols evaluate to zero, thereby preserving the integrity of the target function computation.

- **Channel Model:** To accurately emulate an agricultural deployment, a Channel Model block is introduced to inject realistic wireless impairments. This block models Additive White Gaussian Noise (AWGN) and user-defined multipath delay taps, which scatter the signal arrivals and intentionally induce ISI to test the system's robustness.

3.b.iii Flowgraph Configurations

The network was simulated under two primary conditions. Figure 26 illustrates the continuous-time analog simulation where sensor data is continuously evaluated. Figure 27 depicts the "Faster-Than-Nyquist" (FTN) scenario, wherein the transmission rate is artificially accelerated to break the Nyquist limit and observe the resulting ISI degradation.

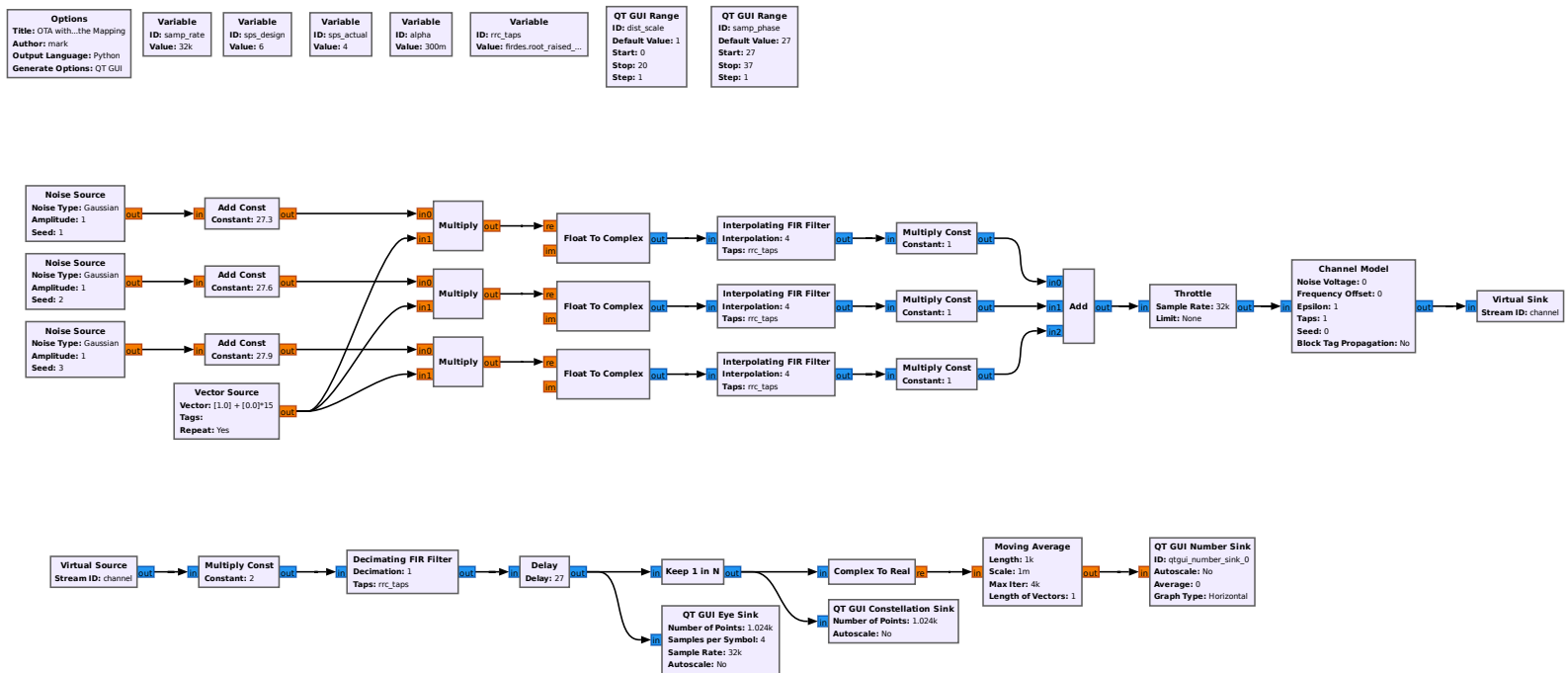


Figure 26: GNU Radio Companion flowgraph for the OTA implementation including Channel Model and RRC filters.

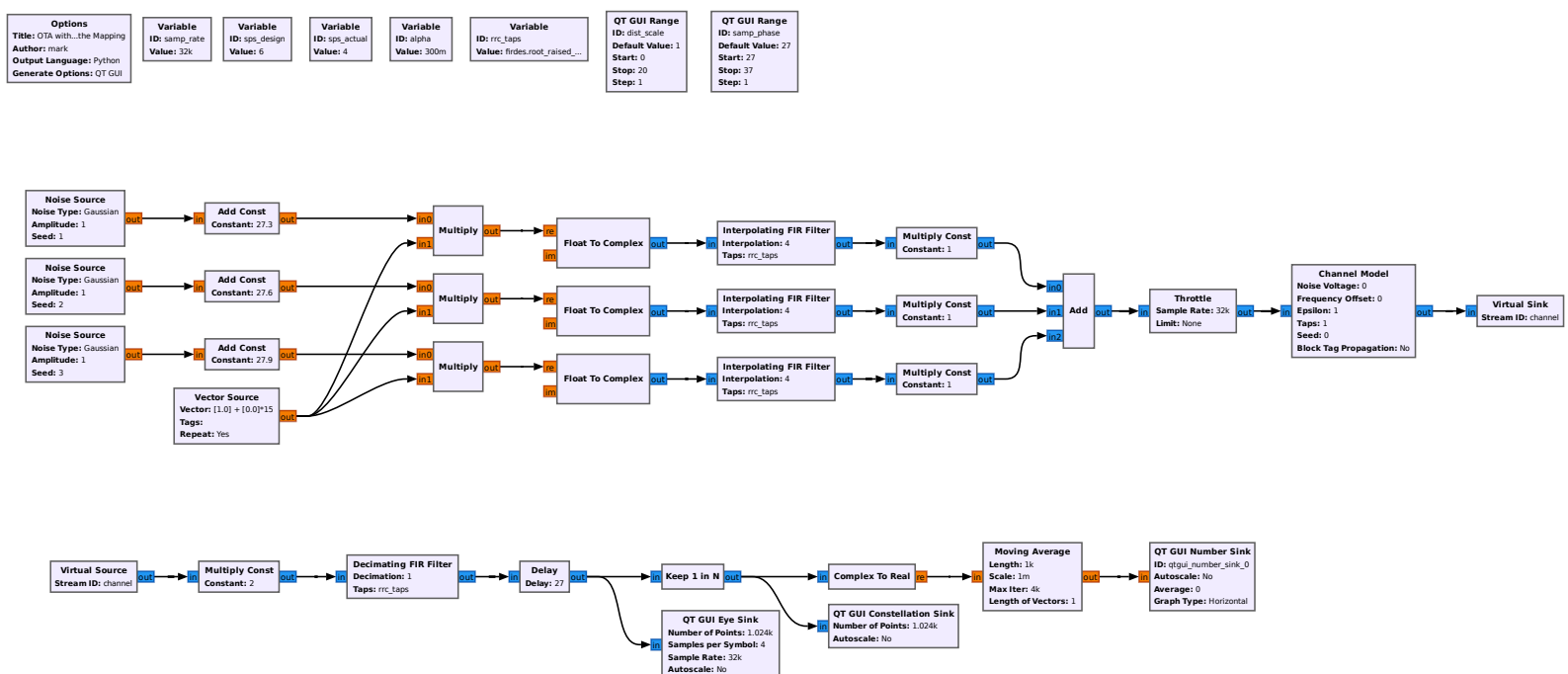


Figure 27: GNU Radio Companion flowgraph configured to induce Intersymbol Interference (ISI) by mismatching the design and actual sampling rates.

3.c MATLAB and Simulink Integration

While GNU Radio provides a robust framework for real-time SDR execution and analog signal evaluation, future phases of this methodology will incorporate MATLAB Simulink. Simulink will be utilized to model advanced feedback loops, custom waveform generation, and digital equalizers (e.g., Minimum Mean Square Error (MMSE) receivers) that are mathematically complex to implement in standard GNU Radio environments. This dual-software approach allows for both practical SDR validation and rigorous mathematical algorithmic design.

4 Experimental Results and Discussion

This section consolidates the empirical and simulated findings of the thesis, structured to provide a comparative analysis of the Over-the-Air (OTA) computation performance across different modeling layers. The discussion first examines the degradation mechanics revealed by MATLAB and Simulink simulations, evaluating the efficacy of applied equalization techniques. It then outlines the evaluation framework for the physical Software-Defined Radio (SDR) deployments.

4.a Comparative Analysis of Channel Impairments (MATLAB vs. Simulink)

The rigorous simulation of OTA architectures across discrete-time algorithmic environments (MATLAB) and continuous-time dynamical environments (Simulink) provided deep insights into the physical limitations of signal superposition.

Under ideal AWGN conditions, both simulation environments confirmed the theoretical Mean Squared Error (MSE) scaling laws, demonstrating that computation variance is strictly bounded by the channel noise [9]. However, the introduction of uncoordinated physical impairments revealed significant divergence from the idealized theory:

- **Multipath and Fractional Delay:** The high-fidelity simulations demonstrated that even sub-symbol fractional delays (modeled via Farrow filters) induce severe Intersymbol Interference (ISI). The 2D and 3D MSE surface scans revealed an irreducible error floor that scales with the multipath gain, fundamentally bottlenecking the computation accuracy regardless of the transmission power.
- **Asynchronous Carrier Frequency Offset (CFO):** Independent local oscillator offsets among distributed transmitters were shown to convert stationary baseband values into beat-frequency harmonics. Simulink evaluations demonstrated that traditional coherent detection fails catastrophically under these conditions. The application of non-coherent Euclidean magnitude detection mitigated the zero-crossing signal collapse but introduced a multi-user vector fading penalty.

4.b Evaluation of Equalization and Recovery Strategies

The simulations also served to benchmark classical equalization and synchronization recovery schemes. The divergence between algorithmic MATLAB models (which assume perfect block-level knowledge) and Simulink models (which emulate real-time causal processing) highlighted the difficulty of correcting asynchronous CFO post-superposition. While techniques such as squaring loops and Costas loops are standard for point-to-point links, their application to aggregated OTA signals proved inadequate when phase offsets are uncorrelated across multiple transmitters. This finding justifies the ongoing investigation into more robust, adaptive filtering mechanisms.

4.c Framework for SDR Experimental Validation

Building upon the simulation baseline, the physical deployment utilizing GNU Radio and SDR hardware aims to validate the theoretical error floors in a real-world electromagnetic environment. The planned experimental results will be benchmarked

against the Simulink diagnostics to quantify hardware-specific non-linearities, such as amplifier saturation and discrete sampling jitter. Ultimately, the experimental framework will demonstrate the practical computation rate achieved by the OTA approach relative to traditional orthogonal access schemes (e.g., TDMA), illustrating the critical balance between resource savings and analog computation error.

5 Conclusion and Future Work

5.a Summary of Thesis Findings

This thesis has systematically investigated the feasibility, theoretical scaling, and physical limitations of Over-the-Air (OTA) computation. By developing a comprehensive cross-platform modeling methodology—spanning mathematical derivation, discrete-time MATLAB simulation, continuous-time Simulink architectures, and Software-Defined Radio (SDR) realization—the research bridged the significant gap between idealized analog computation theory and practical deployment.

The fundamental analysis confirmed that under ideal conditions, OTA computation achieves unprecedented spectral efficiency by exploiting the superposition property of the wireless Multiple-Access Channel. However, the rigorous evaluation of channel impairments demonstrated that the system's performance is severely bottlenecked by uncoordinated transmitter dynamics. Specifically, multipath propagation and fractional timing delays induce Intersymbol Interference (ISI) that manifests as an irreducible error floor. Furthermore, asynchronous Carrier Frequency Offsets (CFO) cause catastrophic coherent demodulation failure due to beat-frequency oscillations. While non-coherent envelope detection provides a robust fallback, it suffers from vector magnitude fading. The findings conclusively show that traditional point-to-point equalization schemes (e.g., standard Costas loops) are insufficient for multi-user aggregated signals, necessitating paradigm shifts in how synchronization and filtering are handled for OTA networks.

5.b Future Research Directions

The insights gained from the physical degradation models open several critical pathways for future research in OTA computation and federated edge learning [10], [11]:

- **Adaptive and Non-Linear Equalization:** Transitioning from static recovery loops to dynamic, adaptive algorithms is paramount. Future work will explore the deployment of Unscented Kalman Filters (UKF) and Extended Kalman Filters (EKF) for the real-time tracking and compensation of non-linear impairments directly from the aggregated waveform.
- **Multi-Antenna and MIMO Systems:** Expanding the single-antenna architecture to Multiple-Input Multiple-Output (MIMO) systems offers spatial degrees of freedom to combat fading. Implementing spatial equalization and receive beamforming [9] could significantly mitigate the destructive interference caused by asynchronous phase offsets.
- **Large-Scale Federated Learning Deployments:** As OTA computation proves to be a natural fit for the aggregation step in Federated Learning, future physical deployments should evaluate the impact of the quantified OTA error floors on the convergence rates of distributed neural network training.

A MATLAB Code for OTA Simulation

This appendix contains the core MATLAB scripts used to generate the simulation data discussed in Chapter 2. Note that we utilized the Raised Cosine method [9].

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